

BrainBay - User Manual

Version 2.8, Jan. 2026

Author: Christoph Veigl, webpage: <http://brainbay.lo-res.at>

Contents:

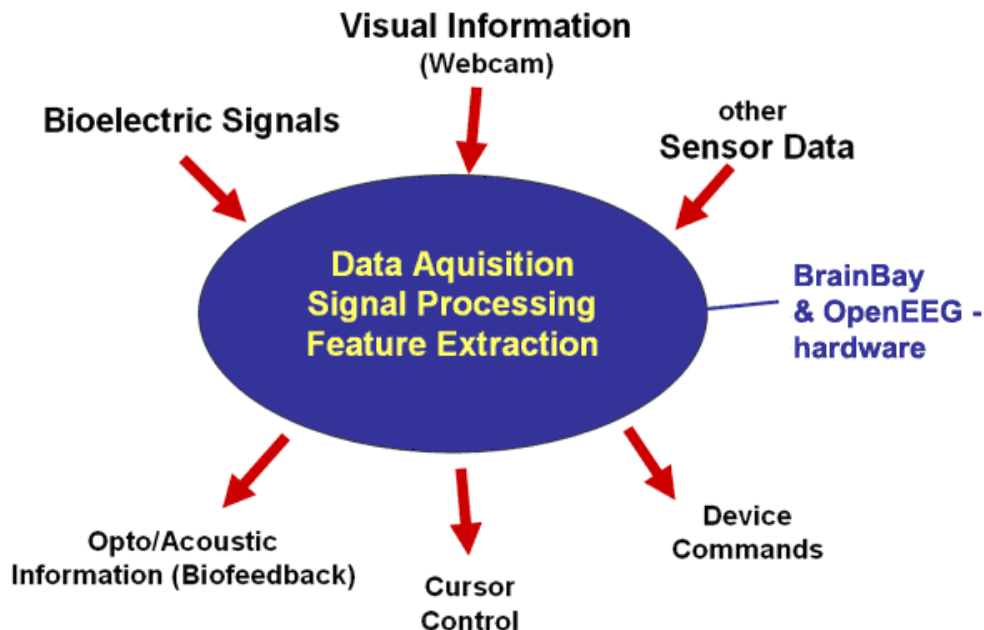
1.	Introduction.....	3
1.1	Possible Applications	4
1.2	Functional features of BrainBay.....	4 [^]
2.	Installation and System Requirements.....	5
2.1	Directory Structure	5
3.	Main Menu and Status Bar.....	6
3.1	Options and Application Settings	7
3.2	Status Bar and Session Control	8
3.3	Tools menu: Color Palette Editor	9
3.4	Tools menu: Tone Scale Editor.....	9
4.	Creating Designs	10
4.1	Inserting, Tagging and Deleting Elements.....	10
4.2	Connecting Elements	10
4.3	Displaying a Connection.....	11
4.4	Setting Signal Ranges and Description	11
5.	Using Design-Elements	12
5.1	Signal Sources	12
5.1.1	EEG- and Biosignal Amplifiers	12
5.1.2	Signal Generator	15
5.1.3	Constant.....	15
5.1.4	EDF-Reader.....	15
5.1.5	File-Read	16
5.1.6	Generic BLE Receiver	17
5.1.7	Generic UDP Receiver	17
5.1.8	Key-Capture.....	18
5.1.9	TCP-Receive	19
5.1.10	Camera (Webcam).....	20
5.1.11	Skindialog.....	21
5.2	Processing Elements	22
5.2.1	Averager	22
5.2.2	Comparator.....	22
5.2.3	Correlation	23
5.2.4	Counter / Display	23
5.2.5	Debounce.....	23
5.2.6	ERP - Detector (Pattern Recognition).....	24
5.2.7	Expression Evaluator.....	25

5.2.8	Filter	25
5.2.9	FFT (spectral analyser).....	26
5.2.10	Integrator.....	27
5.2.11	And, Or and Not	27
5.2.12	Matlab Transfer.....	29
5.2.13	Magnitude	29
5.2.14	Mixer.....	30
5.2.15	Sample and Hold	30
5.2.16	Standard Deviation.....	30
5.2.17	Threshold / Meter	31
5.3	Signal Targets.....	32
5.3.1	AVI – Player	32
5.3.2	Com- Writer	33
5.3.3	EDF- Writer	33
5.3.4	File – Write	34
5.3.5	Oscilloscope.....	35
5.3.1	OSC-Sender	36
5.3.2	Keystroke.....	36
5.3.3	Particle Animation	37
5.3.4	Midi Output	38
5.3.5	Mouse Controller.....	39
5.3.1	Shadow	40
5.3.2	Sound Player	40
5.3.3	TCP – Sender.....	41
5.3.4	Media Player	42
5.3.1	Volume	42
5.4	Other Elements.....	43
5.4.1	Documentation.....	43
5.4.2	Ballgame	43
5.4.3	Button.....	44
5.4.4	Speller	44
5.4.5	Sessiontime	45
5.4.6	SessionManager	46
6.	Example Designs.....	47
6.1	A design for Alpha / Beta – Neurofeedback.....	47
6.2	Heartbeat-Feedback from a multichannel EDF-recording.....	50
6.3	A design for Muscle-Feedback and –Rehabilitation	51
6.4	Controlling a movie using an external Reed-Switch	53
6.5	A Camera - Mouse application with clicking functions.....	54

1.Introduction

This manual provides information and instructions for setup and usage of BrainBay, an Application for Biofeedback, Neurofeedback and Human Computer Interaction.

BrainBay allows the graphical design of universal configurations for realtime-processing, display, storage and opto/acoustic feedback of biosignals, bioelectric events, visual information or other sensor data.

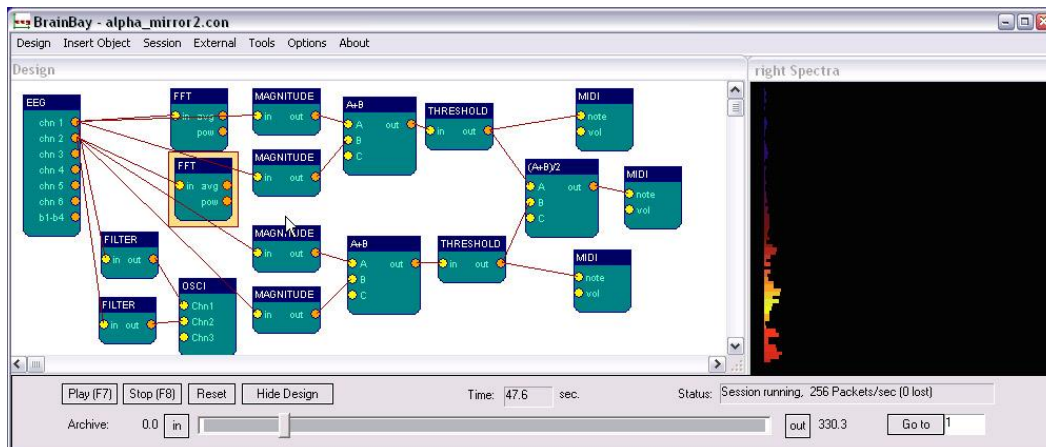


BrainBay is part of the OpenEEG - project (<http://openeeg.sf.net>). It supports the ModularEEG and MonolithEEG biosignal-amplifiers and other affordable off-the-shelf amplifiers. With appropriate electrodes / sensors, following vital parameters could be measured and utilized for biofeedback purposes:

- Brainwaves (Electroencephalogram, EEG)
- Heartbeat (Electrocardiogram, ECG)
- Muscle Activity (Electromyogram, EMG)
- Eye Movements (Electrooculogram, EOG)
- Blood Volume Pressure (BVP)
- Galvanic Skin Response (GSR)
- Skin Temperature
- Accelleration, control switches
- Face-position, body movements (using a webcam)

In a BrainBay *Design-Configuration*, these signals are connected to processing-elements for feature extraction, transformation and visual / acoustic feedback.

The following figure shows a design configuration that connects two channels of an EEG-amplifier to processing- and display-elements that provide visual and acoustic feedback of signal parameters (for details on this design see chapter 6.1) :



1.1 Possible Applications

- Biofeedback- and Neurofeedback-Therapy indications:
tension-headache, migraine, post-traumatic stress syndrome,
tinnitus, epilepsy, chronic pain, cardiac arrhythmia,
concentration- / learning difficulties, incontinence;
- Rehabilitation Training
muscle- workout and relaxation, posture training
- Realtime–Telemetry of Bodysignals
- Patient Monitoring
- Therapy Progress Control
- Online and Offline Data Analysis
- Human Computer Interaction, including:
control of mouse-cursor + clicking-functions,
typing via an On-Screen Keyboard,
Brain Computer Interface

Please note that the OpenEEG-hardware is not certified as a medical device, all application is done at own risk. Biofeedback should be applied by approved therapists and works best as part of a wider psychophysical treatment concept.

1.2 Functional features of BrainBay

- Digital Filters, FFT-Displays (Bar-Graph, Spectrogram, 3d-View)
- Math. Elements (correlation, threshold, trial averaging, expression-evaluator, ...)
- Multimedia feedback (Midi, Wav- and Avi-Playback, Bar Display, ...)
- Network transmission (using the Neuroserver framework)
- Reading and storage of archive-files in European Data Format (EDF)
- Webcam support and face-detection (for head-mouse support)
- HCI-functions (mouse control, keyboard events)

2. Installation and System Requirements

BrainBay uses the Windows operating system (all versions from XP to Win-10). An (incomplete) Linux-build is available, using the Wine-compiler. Using Wine, the standard BrainBay executable can also run under Linux. Minimum hardware requirement is a Pentium-II laptop- or desktop-computer. For more complex designs and/or fast graphic displays, an OpenGL-accelerated graphics-adaptor is recommended.

The installer (executable file) can be downloaded from <http://brainbay.io-res.org>. The binary and data files will be installed in the User\AppData folder. The source code and development files are hosted at GitHub (<https://github.com/ChrisVeigl/BrainBay>)

2.1 Directory Structure

Archives: contains signal archives in P2/P3/EDF format. These archives are used by demo configurations and could be used for own experiments. This is the recommended folder for storing own archive files.

Configurations: contains design-configurations (.con - files). The delivered design-configurations shall demonstrate main functions of the application

ComputerVision: contains data for the facetracker (camera) element

Graphics: contains images for the particle generator or other processing elements

Movies: contains videos for the avi-player element (.avi files)

Patterns: contains recorded patterns for the pattern-recognition element (.erp files)

Palettes: contains sample color palettes for display elements (.pal files)

Reports: contains oscilloscope screenshot bitmaps (.bmp files) or ASCII-archives (.cvs files)

Tonescales: contains sample tonescales for the Midi-Player (.sc - files)

Network: contains executables for the *Neuroserver* software framework. (<http://openeeg.sourceforge.net/doc/software/NeuroServer>)
These files are needed to transmit or receive edf-files via TCP/network.

Sounds: contains audio files for the audio-player element (.wav files)

Skindialogs: contains a sample skindialog for user interaction (.ini and .bmp - files)

3. Main Menu and Status Bar

All main functions of BrainBay can be accessed through the menu on top and the status-bar on bottom of the window. The menu provides the following entries:

Design: create a new design, load / save a design, exit application

Edit: copy / paste an element (can be used to paste to another BrainBay instance)

Insert Element: add new elements to the design

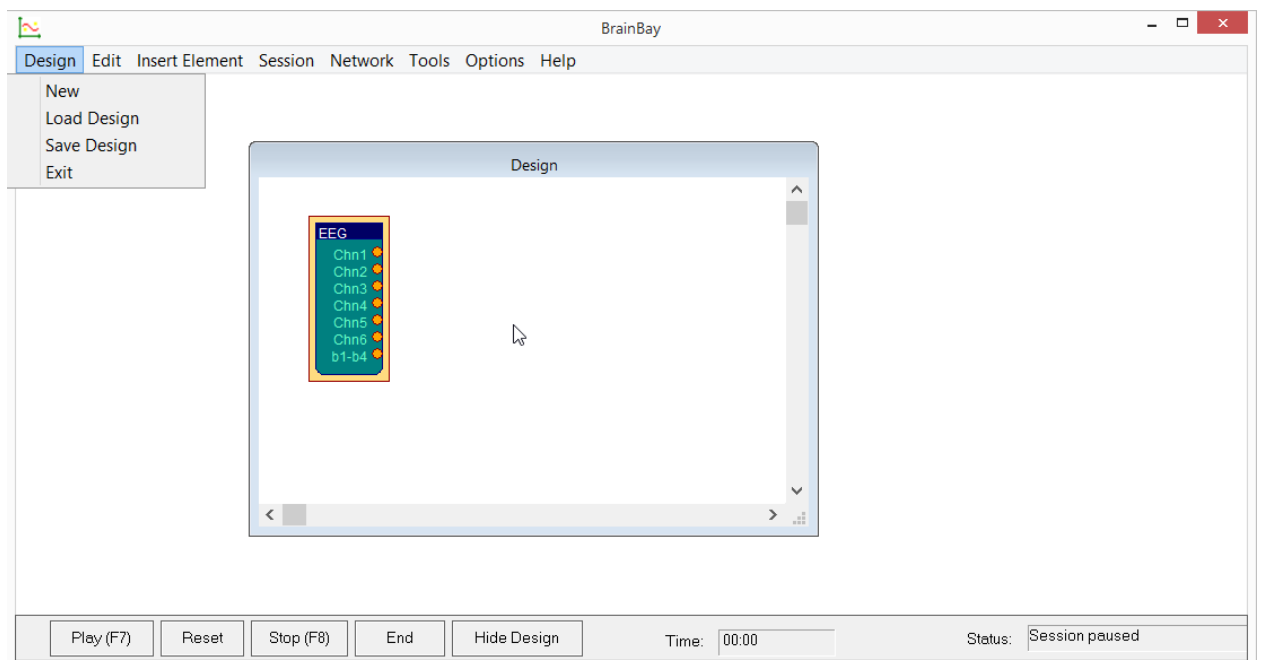
Session: run or stop the designed session

Network: start a neuroserver - service for network transmission of biosignal data or an EDF-reader client, these are part or the neuroserver-framework.
for more information on network transmission see the TCP-read / -write -elements (..)

Tools: editors for color-pallettes and midi-tonescales

Options: settings for application and currently active eeg amplifier (see 3.1)

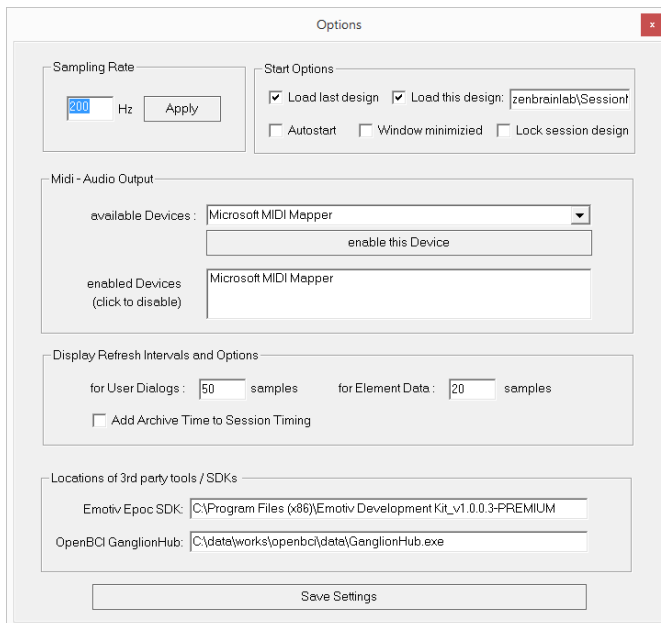
About: program-version and author information



3.1 Options and Application Settings

The Main Menu's *Options* – section offers two dialogs: the Application Settings and the Device Settings. The Device Settings will only be accessible if an EEG amplifiers has been added to the Design. The Application Settings offer several important parameters for system configuration and startup.

Application Settings Dialog:



In section *Sampling Rate*, the sampling frequency (which also determines the pace of signal processing) can be selected. This value should match the sampling rate of the data acquisition hardware or recorded archive file.

For example, the ModularEEG, sampling rate will usually be 256Hz. MonolithEEG provides sampling rates up to 1200 Hz. When reading EDF-Archive files with other sampling rates, the value has to be adjusted accordingly.

The *Start Options* section has options for application-initialisation: If *Load last design* is chosen, the previously used design will be loaded automatically. *Load this design* allows to specify a name of a particular design configuration which should be loaded (the file name is given without the .con extension and may contain a relative path from the *Configurations* subfolder). If *Autostart* is enabled, the session starts after being loaded. *Window minimized* results in automatic minimisation of the main window after a session has been loaded. The *Lock session design* checkbox allows protection the design configuration against unintended changes by users: if this option is enabled, GUI windows can not be re-arranged and the *ShowDesign* button will be hidden from the status bar (note that the *F5* function key will still work to display the design, see 3.2).

The *Midi-Audio Output* section defines the Midi-Devices that will be active within BrainBay. A Midi-Device could be a soundcard with midi-synthesizer, a wavetable software synthesizer, or external Midi-devices like keyboards, drum-computers or midi-controllers. Virtual Midi device drivers like the *LoopBe1* freeware create virtual Midi-input and -output devices that provide a bridge to other midi-processing software running in background.

To add a Midi-Out-Device to the list, select it from the combo-box which contains all devices that are currently accessible in the system. To delete a device from the list, just click the list-entry.

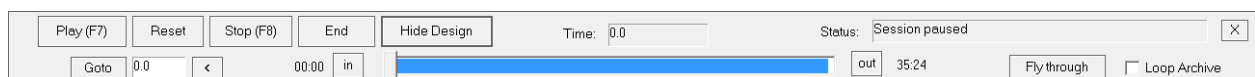
In section *Refresh Intervals*, the update rates for screen displays are set. *User Dialogs* selects the refresh rate for element's parameter windows, *Display Windows* selects drawing update rates for oscilloscopes, FFTs or meter windows. The refresh interval is given in relation to the sampling rate: a value of 8 gives 32 updates per second when a sampling rate of 256 is used.

In the section *Locations of 3rd party tools / SDKs*, path settings to additional resources for specific EEG amplifiers can be defined:

- The *Emotiv Epoc SDK path* setting is only necessary if you want to interface the Emotiv Epoc device via the (non-free) SDK for accessing Epoc's raw EEG data-stream. In this case, specify the correct path to the SDK (where the edk.dll file is located) so that the emotiv eeg element can be used.
- The *OpenBCI Ganglion Hub* is only necessary if you want to use the OpenBCI ganglion EEG device. In this case, specify the path to the GanglionHub.exe application which provides the raw data from the ganglion device. The GanglionHub is delivered as part of the OpenBCI GUI or can be downloaded separately from the OpenBCI github page.

3.2 Status Bar and Session Control

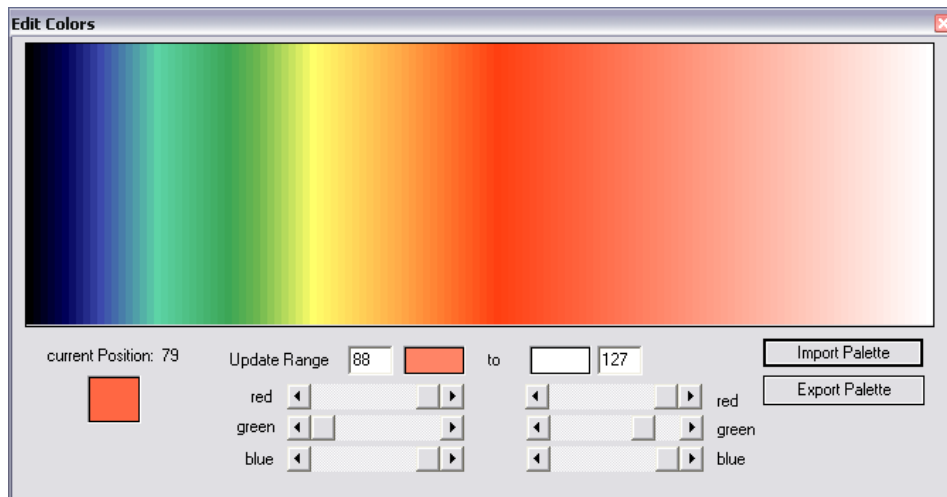
The status bar at the bottom of the main window provides various information (current session time, status field) and buttons for session control. The *Play* and *Stop* - buttons or the respective hotkeys *F7* and *F8* can be used to start and stop the session. *Reset* sets the position of archive files to their beginning and the session time to 0 seconds. The *End* Button halts a currently running session and loads the design configuration which has been specified in the *Application Settings* dialog (parameter *Load This Design*, see 3.1). The *Hide/Show Design* - button or hotkey *F5* controls visibility of the design-window. The status bar itself can be hidden or displayed by pressing *F6*.



If there are archive-file readers present in the design configuration, the status bar extends automatically and shows the progress of the playback. In- and out points can be defined for the archive playback which is useful when particular parts of the recording shall be reviewed in more detail (see description of the EEG-amplifier and EDF-reader elements). The *Loop Archive* option allows to automatically reset the archive playback position to the beginning when the end has been reached (endless playback). The *Fly through* button processes the archive at maximum speed (this is useful to review overall session outcomes without having to wait until the real time playback finishes).

3.3 Tools menu: Color Palette Editor

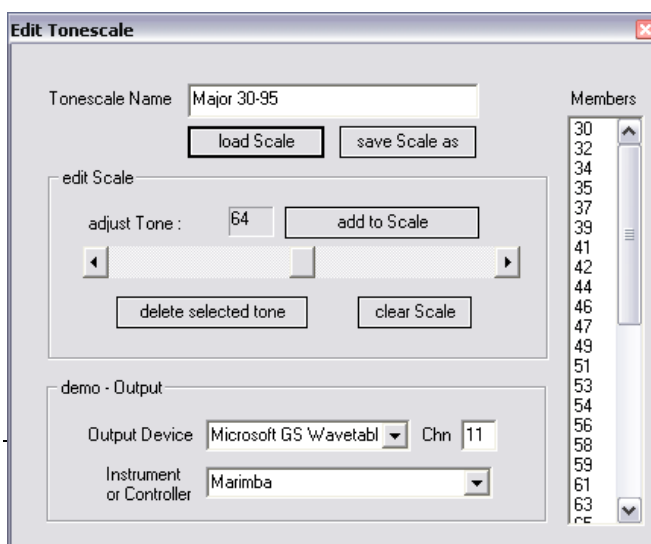
The *Color Palette Editor* enables to easily create color fades and palettes for some display elements. Currently, the color palettes are used by the *FFT* spectral analyser and the *Particle Animator* elements.



To adjust the colors, use left- and right clicks into the color-window to set the update region: A left-click sets the start and a right click the end point of the current region. Using the smaller color selection fields or the scrollbars for RGB - composition, the color value of the start or end point can be set and a color fade from start to end is calculated. Thus, smaller regions or one big region can be created per just a few clicks. A palette consist of 127 colors. Existing palettes can be load and modified using the *Import Palette* button, new palettes can be saved using *Export Palette*.

3.4 Tools menu: Tone Scale Editor

The *Tone Scale Editor* allows the creation own Tonescales for the midi-player element like major, minor, blues etc. Sometimes it can be useful to make small scales with just two or three tones that fit together.



The current tone is selected with the scollbar, it will be played using the given midi-output device, channel and instrument. By pressing *add to Scale*, the tone is appended to the members of the tonescale.

To play the whole or parts of the scale, select a tone from the list of member and use the cursor up/down to browse the scale.

Use *Load Scale* and *Save Scale* to modify existing scales or to create new ones.

4. Creating Designs

The process of creating design configurations with BrainBay basically consists of the following steps:

- Insert elements to the design window
- Connect signals from an element's output port to another elements input-port
- Adjust properties in the element's graphical user dialog
- Test the design with archive-, simulation- or realtime data

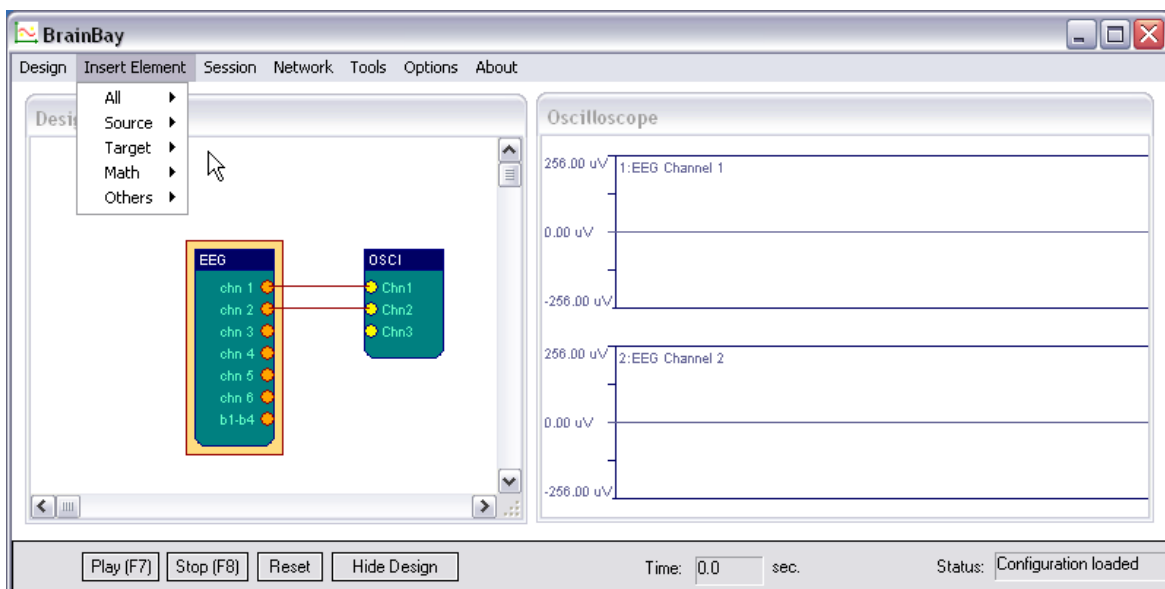
4.1 Inserting, Tagging and Deleting Elements

The *Insert Element* - entry of the main menu shows up 4 groups :

Signal Sources, Signal Targets, Processing - Elements and *Others*;

Generally, every reasonable design will need at least one signal source (like an EEG-Amplifier, a signal generator or a file-reader) and one signal target (like a spectral-display or an oscilloscope).

An element can be selected by left-clicking the center. The active element is drawn with a yellow border. Connections can only be drawn from the active element to other elements. The active element can be deleted with the *Del* - Key of the keyboard. By double-clicking the center of an element, its description-tag can be modified.



4.2 Connecting Elements

A connection is drawn by left-clicking an output-port of the active element and dragging the line to a free input-port of another element.

Signal-sources have at least one *Output Port* (orange). The above picture shows 6 output ports on the EEG-element for the channels of the EEG-amplifier and a seventh output port for the status of control switches.

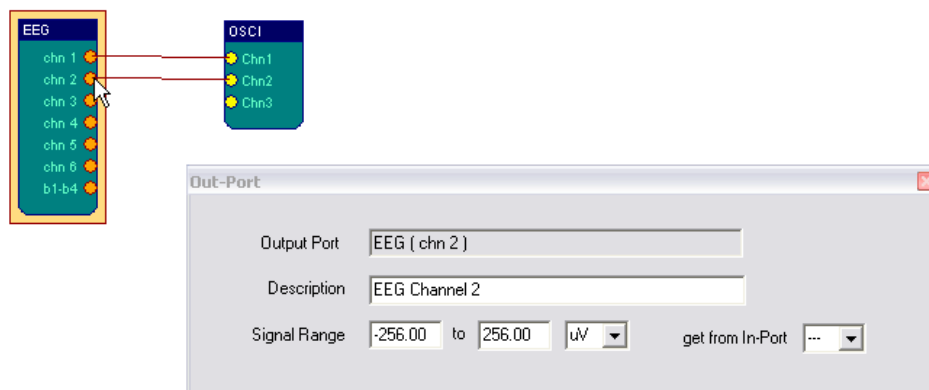
Signal-targets have at least one *Input Port* (yellow).

4.3 Displaying a Connection



By left-clicking a connection, a dialog with information about the connection is displayed. It states the connected ports, a description of the signal and the signal-range.

4.4 Setting Signal Ranges and Description

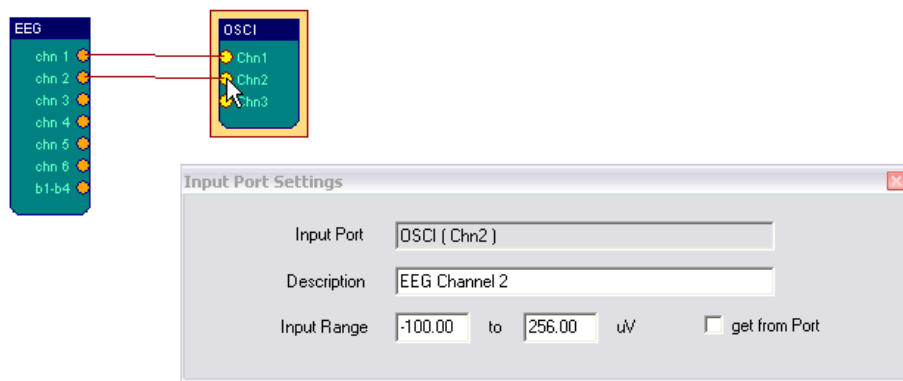


The signal-range and description of a connection can be adjusted by right-clicking an input- or output port. This opens the *Input-Port-Settings* or respectively the *Output-Port-Settings* dialog.

The description for the signal can be written into the given field.

The above example shows how to adjust the signal range of *Chn2* output port of an EEG-signal source element. For elements which have input ports, the signal range of an output port can be obtained from a selectable input port. Thus, signal-ranges can be propagated to following elements.

The input-range for a specific input port can be adjusted manually or inherited from the connected output port. Adjusting the input-range can be useful for elements which define minimum/maximum settings for their functional behaviour according to the input port range (e.g. Oscilloscope display or Midi tone range).



The *Input-Port Dialog* for the oscilloscope-element (here: channel 2).

Note: Brainbay uses a special signal value to indicate “invalid” or “out-of-range” signals – this value is called `INVALID_VALUE` and is represented by a constant (-32767). See more details in the description of the logical elements *And*, *Or* and *Not*.

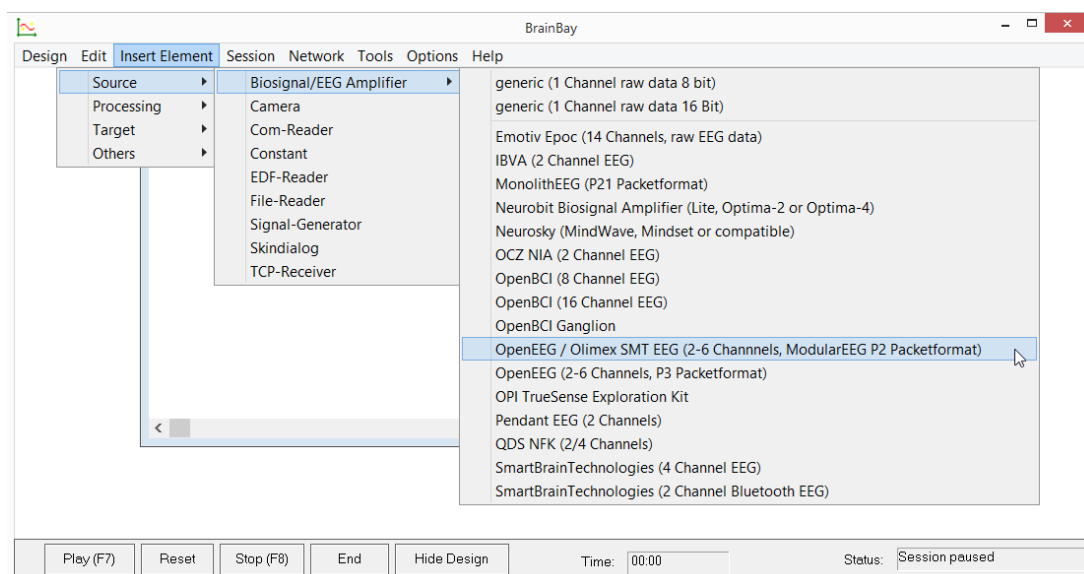
5. Using Design-Elements

Design-Elements (Signal Sources, Signal Targets and Processing elements) are the fundamental components for building a signal processing configuration in BrainBay. Nearly all elements have a graphical user-dialog (configuration window) to adjust their parameters. The user-dialog can be displayed by right-clicking the center of an element. In the following, the BrainBay design-elements and their usage will be described.

5.1 Signal Sources

5.1.1 EEG- and Biosignal Amplifiers

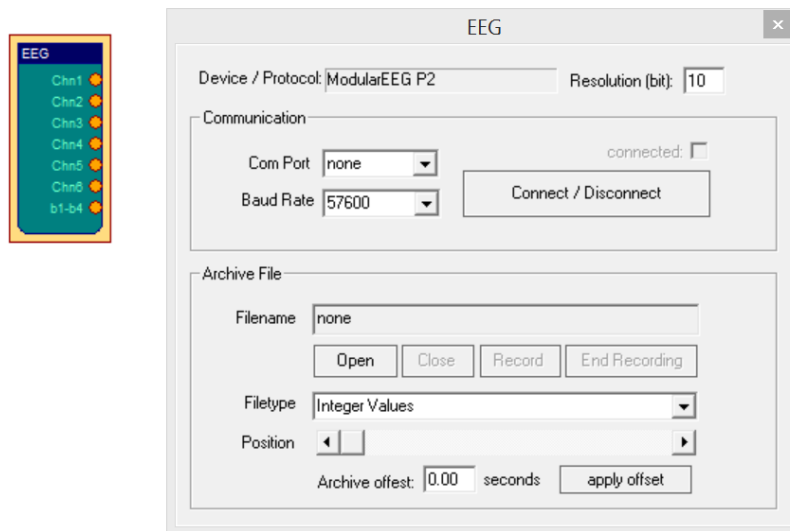
The supported EEG- and Biosignal Amplifiers are available in a subgroup of the Signal Source Menu. Currently, 14 different devices / protocols are supported:



For some of the biosignal amplifier devices, different firmwares and transmission protocols exist, for example the P2, P3 and P21 firmware variants for the OpenEEG device. Some of the protocols are bi-directional (which means that values can also be sent to the amplifier). For example, the MonolithEEG P21 is a bi-directional protocol which allows changing the baud- or samplingrate, setting the active channels and transmitting play/stop-commands to the device.

Additionally to providing live-data from amplifier-hardware, many of the EEG elements can record and playback archive-files: The element either works in live- or in archive mode (not both at the same time).

The following screen shot shows the OpenEEG / Olimex SMT EEG device window:



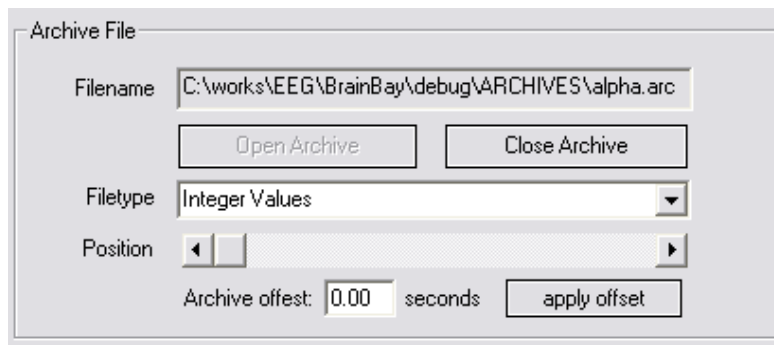
Receiving live-data from the amplifier

Adjust the samplingrate for your recording device in *Options->Application Settings*. Connect the amplifier to a USB-Port (or perform the necessary steps to pair the wireless connection via Bluetooth) and select the appropriate Com-Port from the combo-box in the EEG element properties. Select the baud-rate for the device and adjust the resolution (if necessary). Press button *Connect/Disconnect Com Port*. The *connected*-checkbox should indicate that the port has been connected (if not, check Com-Port and settings).

To display live-data, connect an oscilloscope-element to the output ports of the amplifier-element, press *Play* - button from the status bar and turn on your biosignal amplifier.

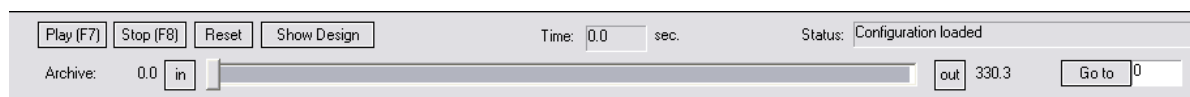
Recording live-data to an archive file:

Setup the connection to the amplifier as described above. Press button *Record Archive* and choose a filename and folder to store the archive. Recording starts immediately after confirming the filename (and starting the design). To stop the recording and close the file, press the button *Close Recording*.



Playback of recorded archives:

Close an eventually opened Com-Port and press the button *Open Archive*. A dialog will appear where an archive file can be selected. The device- and sampling-rate settings will be modified according to the archive. When an archive file is opened, no live-data can be processed by the EEG-amplifier element, and the status bar extends: a slider shows the progress of playback and can be used for positioning in the archive file. The current position in seconds is shown in the *Time* -Field. Use the *Play*-, *Stop*- and *Reset*- buttons to control playback.



With the *In*- and *Out*- buttons, a segment of the archive can be defined for looped-playback. By pressing *In* or *Out*, the current position will be set as the new start or end - point for playback. The *Go to* - button can be used for jumping to a specified position, given in seconds.

Only one *EEG-Amplifier* signal source can be active in a given design. Nonetheless it is possible to have more than one archive players active in the system, using the EDF-File Reader element. To align the position of an EEG-amplifier archive relatively to other archives, an offset for the archive in seconds can be set in the user dialog.

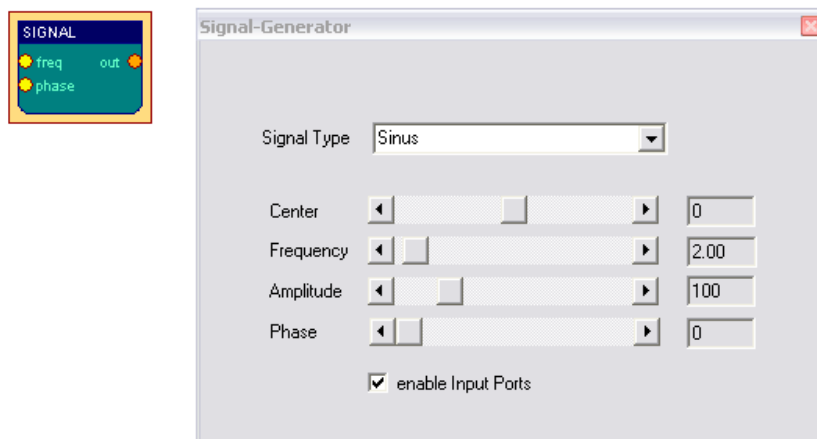
When the *EEG-Amplifier* element receives live-data from an amplifier, the sampling rate for the whole system is controlled by the output of the EEG-amplifier element. That means: when the amplifier sends 512 Packets per second, all other elements will process data at that pace.

When playing back an archive, the sampling rate will be set according to the rate of this archive, but the rate can be changed using the *Options* - dialog.

5.1.2 Signal Generator

The *Signal-Generator* element can be used to produce sinus-, rectangle or sawtooth-signals with selectable center, frequency, amplitude and phase-shift.

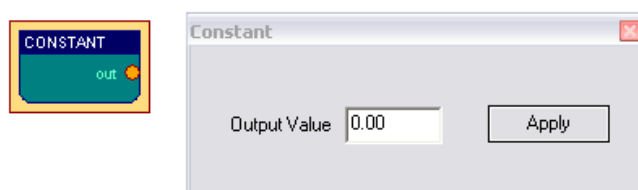
The signal type is selected via combo-box, the other parameters can be changed with the respective silders. The maximum amplitude (the range of the amplitude - slider) can be changed by right-clicking the output port of the element.



The checkbox *enable Input Ports* controls the accessibility of frequency- and phase-settings via input ports. Using the input ports, these settings can be modified by other elements or by a user dialog.

5.1.3 Constant

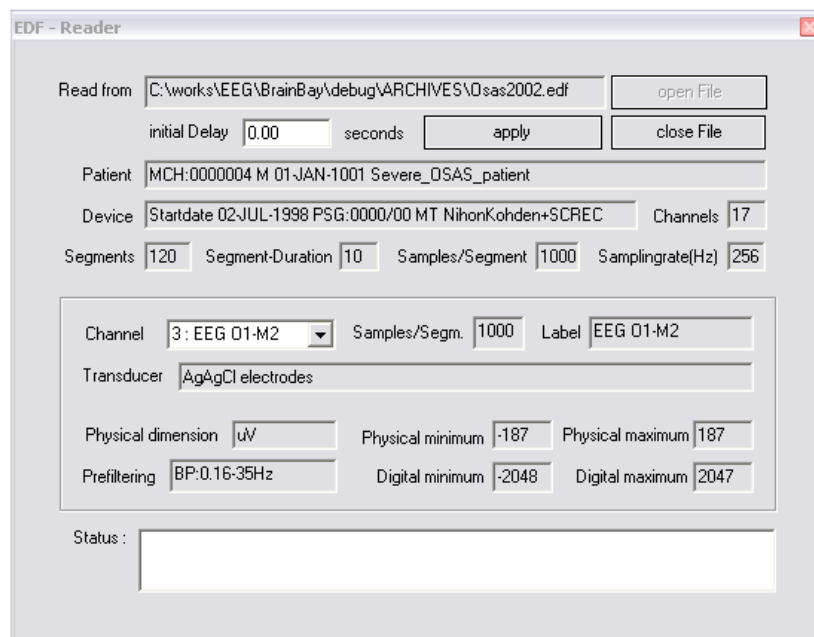
The *Constant* element provides a fixed value on it's output port :



5.1.4 EDF-Reader

The *EDF-Reader* element can be used to read previously recorded file in European Data Format. EDF is a format for the exchange for biological signals (see <http://www.edfplus.info>). The header of an EDF file stores various information about the recording: date, patient data, recording device, sampling rate, data segments, signal ranges and descriptions for the data channels.

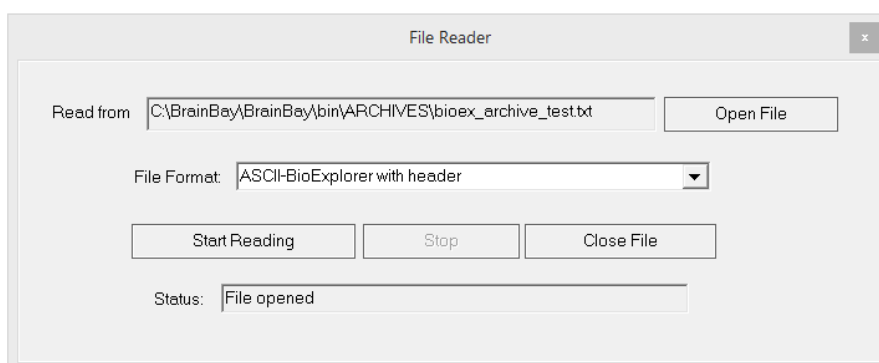
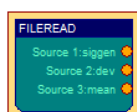
This information is displayed in the user dialog of the *EDF-File Reader* element after a valid file has been opened.



Choose a channel from the combo box to display its information. EDF-files can have many channels. However, BrainBay's channel limit is set to 32. The signal range of the output ports is set by *Physical Minimum* to *Physical Maximum* of the corresponding channel. Similar to the *EEG-Amplifier* archive playback, the status bar extends when an EDF-archive has been opened successfully, to enable playback control and positioning in the archive. (see *EEG-Amplifier* element). The field *Initial delay* allows specifying an offset to align this archive file with other files for playback.

5.1.5 File-Read

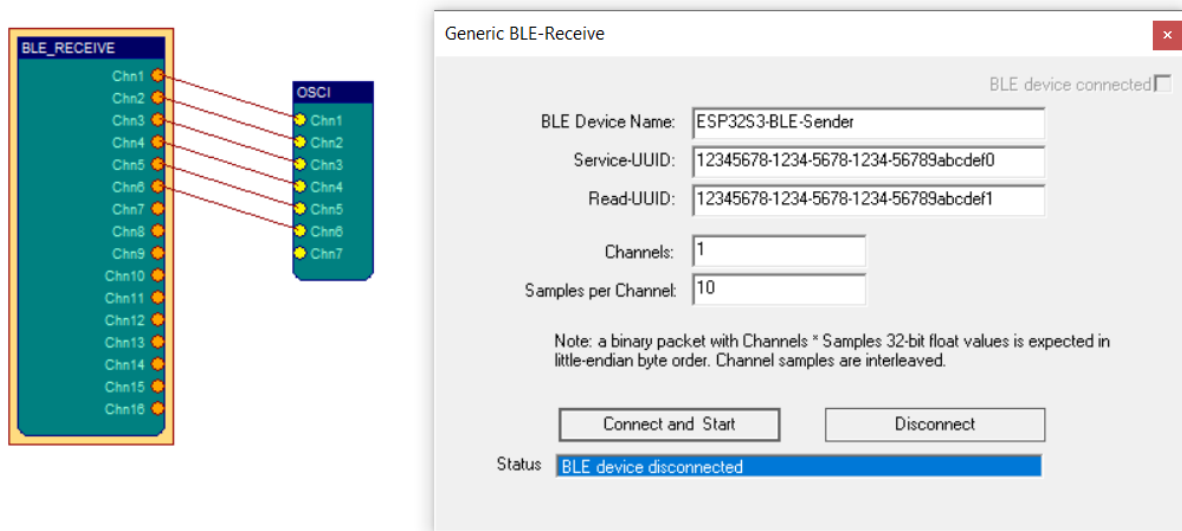
The *FileRead* element allows reading signal channel data from ASCII- text files or raw binary files. It is the counterpart to the *FileWrite* element (data recorded via the *FileWrite* element can be replayed using the *FileRead* element)



Using the *File Format* combo box, the type of storage and the delimiters for rows and columns can be selected. Available formats are: *ASCII-Integer Values* (human-readable text file) *Bioexplorer File Format* (including a header which makes this file compatible to the Bioexplorer application) or *1-channel 16 bit raw* format (see also *FileWrite* element). The archive file read can be started and closed manually via the dedicated buttons in the configuration dialog.

5.1.6 Generic BLE Receiver

The *GenericBleReceiver* element allows connection of devices which send signal data over a BLE connection. The device is identified via its BLE advertising name. Furthermore, a Service-UUID and a Read Characteristic-UUID must be specified to establish the GATT connection. Usually, a compatible BLE sender can be implemented easily using a BLE-capable microcontroller (ESP32, RP Pico W) and the NimBLE library.



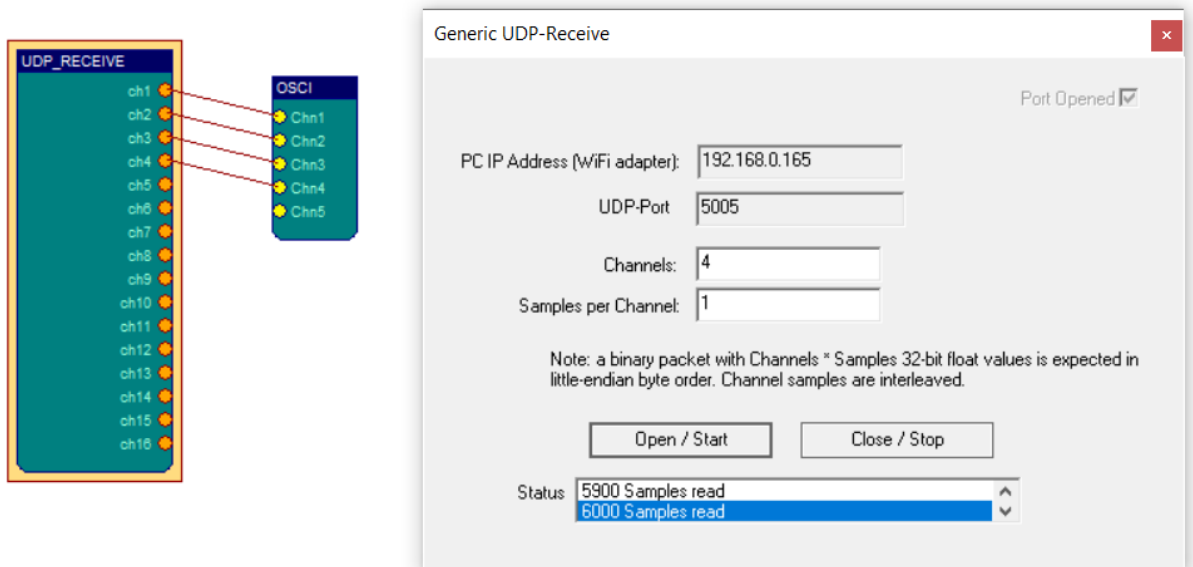
Next to the device identification parameters, two key parameters must be specified to define the structure of the expected data packets which are sent from the BLE device: the number of *Channels* and the number of *Samples*. Note that the BLE packet is expected to contain *Channels* * *Samples* single precision (32-bit) floating point values in little-endian format. The channels are expected to be interleaved. For example, using the settings *Channels*: 3 and *Samples per Channel*: 2, the data packet should be composed as:

Ch1-Sample1, Ch2-Sample1, Ch3-Sample1, Ch1-Sample2, Ch2-Sample2, Ch3-Sample2

which gives a total of 6 float values (in a 24-byte binary data packet). Thus, it is possible to keep the protocol overhead minimal and still allow an arbitrary number of channels and multiple samples in one packet, which is essential for keeping data transfers efficient, especially when using higher sampling rates.

5.1.7 Generic UDP Receiver

The *GenericUdpReceiver* element allows connection of signal senders that send data to a local UDP port. This could be a WiFi-capable microcontroller (ESP32, RP Pico W) connected to the same network. In the configuration dialog, the local IP address and the provided port is displayed – this information is needed for configuration of the UDP sender side. The element searches for local WiFi adapters and provides the associated IP address if possible. If no Wifi adapter is available, the IP address of the wired LAN device is provided.

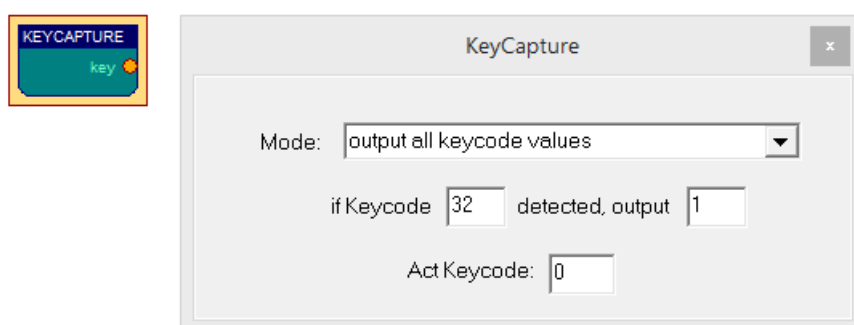


Regarding the expected data packet format, please refer to the description of the Generic BLE receiver element above, which uses the same format.

Note that UDP is a “connectionless” transport protocol without error correction – this means that samples might get lost if connection problems occur, but latency is kept low.

5.1.8 Key-Capture

The *KeyCapture* element outputs information about currently pressed keyboard keys. This can be useful to store triggers in the recording of a biosignal archive (for example pressing the spacebar key when starting meditation or deep breathing) – this information can be added to an EDF file or CSV file recording.



The *KeyCapture* element offers two modes: “*output all keycode values*” sends the keycode of any pressed key (for example 32 for the spacebar key), whereas “*detect one keycode and output custom value*” outputs an adjustable value only if a particular key is pressed – the keycode of the key and the desired value can be given in the respective edit fields. If no key is pressed, the element outputs zero. During a running session, the keycodes of currently pressed keys can be viewed in the “*Act Keycode*” field. Please note that the *KeyCapture* element only works if the main application window has the focus (is activated).

5.1.9 TCP-Receive

Similar to the *EDF-Reader* element, the *TCP-Receive* element provides EDF channel data to other elements. The difference is that the data comes from a network connection and not from a local file. The *TCP-Receive* element connects to the neuroserver software framework by Rudi Cilibrasi which is also part of the OpenEEG project (see <http://openeeg.sourceforge.net/doc/sw/NeuroServer>). To use the TCP-Receive element, a neuroserver has to be running on a known host in the network or on the local host. Sending- and receiving clients can connect to this service. Thus, clients can send / receive live biosignal data and recorded EDF-archives to / from the neuroserver.

```

c:\works\leeg\brainbay\debug\NETWORK\insd.exe
NSD (NeuroServer Daemon) v. 0.7.4-Windows
Binding network socket at 8336
Socket bound.
Please start the modeegdriver.
Accepting from 00000074
Received new connection from 127.0.0.1
Got connection on client 0.
Got header:
The header is <0
MCH:0000004 M 01-JAN-1001 Severe_OSAS_patient
Startdate 02-JUL-1998 PSG:0000/00 MT NihonKohden+SCREC
15.00.0000.30.014608
EG Fp2-M1 120 1 EEG Fpz-M2 EEG C3-M2 EEG O1-M2 E
OG Pos18-M1 EEG C4-M1 EEG O2-M1 EEG M2-M1 EOG Pos8-M1 E
esp chest Resp abdomen SaO2 Hypnogram AgAgCl electrodes
AgAgCl electrodes
AgAgCl electrodes
AgAgCl electrodes
AgAgCl electrodes
AgAgCl electrodes
AgAgCl electrodes
AgAgCl electrodes
AgAgCl electrodes
AgAgCl electrodes
AgAgCl electrodes

```

An easy way to test the *TCP-Receive* element is to start a neuroserver and an EDF-reading client on the local host (by using the *Network-* entry from the main menu) The figure shows the running service, with an EDF-client connected.



TCP-Receive

Neuroserver: localhost Biosignal-Session: 0:EEG

Patient: MCH:0000004 M 01-JAN-1001 Severe_OSAS_patient

Device: Startdate 02-JUL-1998 PSG:0000/00 MT NihonKohden+SCREC Channels: 17

Segments: 120 Segment-Duration: 1 Samples/Segment: 100 Samplingrate(Hz): 256

Channel: 1: EEG Fpz-M2 Label: EEG Fpz-M2

Transducer: AgAgCl electrodes

Physical dimension: uV Physical minimum: -180 Physical maximum: 180

Prefiltering: BP:0.16-35Hz Digital minimum: -2048 Digital maximum: 2047

Status: clear OK OK

lost packets:

Connecting the TCP-Receive element to the neuroserver service

Enter the IP-adress of the host computer where the service is running into the field *Neuroserver* of the element's user dialog. (when running a local service type 'localhost'). Press the *Connect* - button. In case of a successful connection, the *Status*-listbox will show the line 'OK' and the combo-box *Biosignal-Session* will be filled with the available sessions from the neuroserver. Choosing one of the sessions will open the EDF-stream, display the header information and update the element's output

ports. The port ranges for the output ports are obtained from the channel-information of the EDF-file.

During data reception, the progress in packets is shown in the *Status*-listbox of the user dialog. Use the *Stop* -button to bypass reception and *Close* to disconnect from the neuroserver.

Please take into account that there will be some delay in the transmission, depending on network bandwidth and other factors like the receive buffer for incoming values. The button *Empty Receive-Buffer* discards packets that are currently buffered

5.1.10 Camera (Webcam)

The Camera element can be used to process a video stream from a local camera (usually a webcam) and perform face- and movement-detection. For image- processing, the *Intel Open Computer Vision library* was used. (*OpenCV*, see <http://www.intel.com/research/mrl/research/opencv>)

In particular, three tasks can be performed by the Camera element:

- Record a live-videostream to an .avi - file to harddisk
- Playback of an .avi - file from harddisk
- Use a face detection - and feature tracking algorithm to output the estimated position of the user's nose and chin. Together with the *Mouse Controller* - element, this information can be used to control the cursor and clicking.



To use the camera element, a webcam or another camera needs to be connected and visible to the system (installation of correct device drivers etc). After inserting the camera element into the design, the device is opened and the live-stream is displayed in a separate window.

The capture interval can be set using the field *Frame Interval* of the element's user dialog. The interval has to be given in milliseconds, and should not be longer than about 65 milliseconds to get 15 frames per second. The value strongly depends on the webcam used and the overall system performance. Using a good webcam, 25 frames per second can be captured using a 1,6 Ghz machine with USB 2.0.

The button *Display Camera Settings* calls the settings dialog of the webcam. There, camera-specific parameters like contrast or the frame rate can be adjusted.

Three radio-buttons control the mode for the Camera-element:

'No archive operation' is usually used for feature tracking (head-mouse designs).

When 'Record to archive' is pressed, an open-file dialog can be used to choose a filename and folder for the avi-file. Then, a dialog for codec-selection appears. The codec compresses the image frames before writing them to disk. Choose a codec that does not use much processor time ('uncompressed' is surely the fastest, but gives very large video archives, 'Cinepak' or 'Microsoft video1' are possible alternatives). After selecting the codec and starting the design, the video frames are written to the file until the *Stop Camera* - button is pressed, which also closes the archive file.

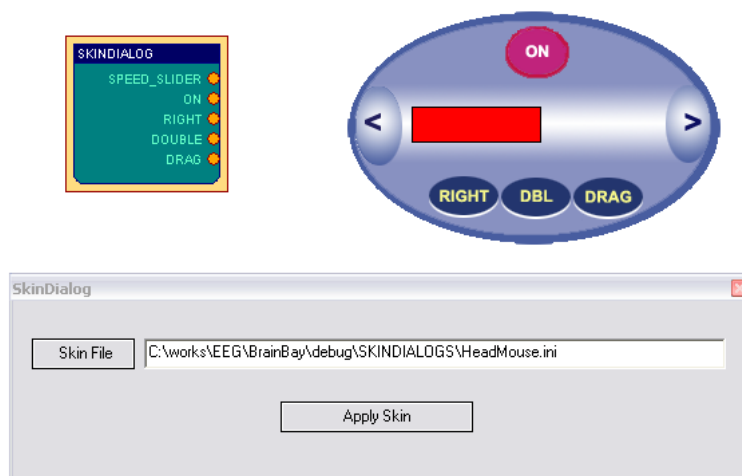
'Playback from archive' also displays an open-file dialog where the .avi file of a previous recording can be chosen. After starting the design, the video stream will be delivered from the archive and not from the webcam. This is also possible when no webcam is installed in the system.

Feature Tracking

If the switchbutton 'enable tracking' is checked, the element will perform a Haar-Cascade detection of a frontal human face on the videostream from camera or video archive file. When a face has been found, the nose and chin positions are estimated and followed by an optical flow algorithm. The coordinates of nose and chin relative to the last image are sent to the element's output port. After appropriate filter- and resize-operations, these coordinates can be used by the *Mouse-Controller* element to perform mouse movement and clicking operations (for details see the Camera-Mouse example design)

5.1.11 Skindialog

The Skindialog Element provides a custom draw, skinned user dialog to the design. Buttons and sliders with mouse-sensitive areas can be defined using an .ini -file and bitmap graphics. The current values of the sliders or button states are presented at the element's output port. Thus, special dialog can be drawn to support users in the access or key features of a complicated design.



The description of the Skinned Dialog is stored in an .ini-File which contains filenames of bitmap-graphics for the activated and deactivated dialog

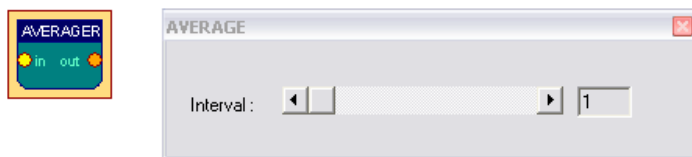
and the pressed buttons. (see the `skin_readme.txt` file in subdirectory *Skindialogs* for more details)

The Skindialog object is currently in beta-stage, a more comfortable editor will be developed that allows online-adjustment of the dialog without editing the `.ini` file.

5.2 Processing Elements

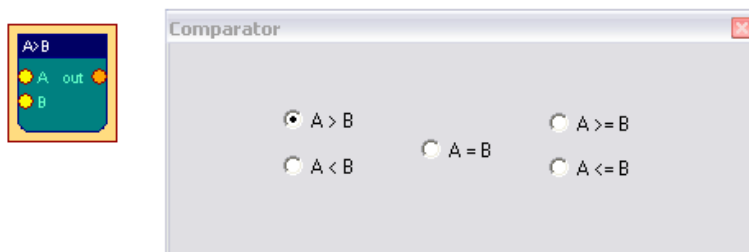
As the signal source elements deliver biological data, simulated waveforms and other signals, the processing elements to perform signal adjustments, transformations, apply thresholds or combine signals in order to get the desired parameters for the bio-feedback and training processes. BrainBay currently has 19 signal processing elements which will be introduced in the following.

5.2.1 Averager



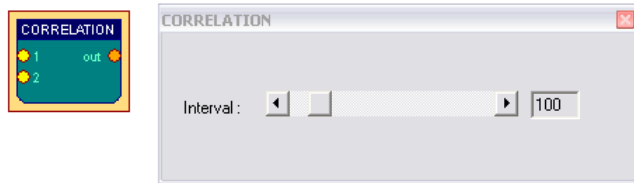
The *Averager* element puts out the mathematical average of the last n samples. The number of samples can be selected by the Interval scroll bar.

5.2.2 Comparator



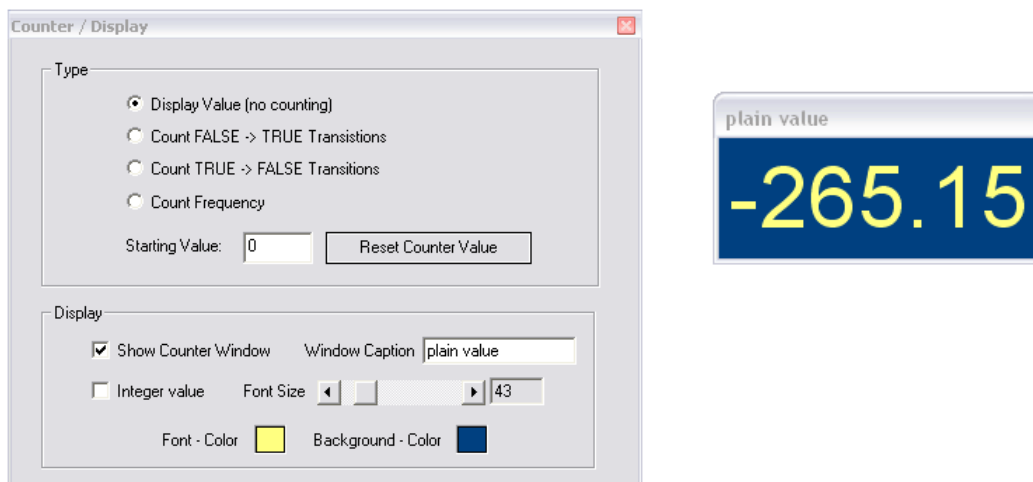
This element compares its two input values in a way given by the user. When the selected condition is fulfilled, the value of input A is routed to the element's output port. If not, the output of the element will be `INVALID_VALUE` (this is a representation of boolean false). In case one of the inputs is `INVALID_VALUE`, the output of the element will be `INVALID_VALUE` as well.

5.2.3 Correlation



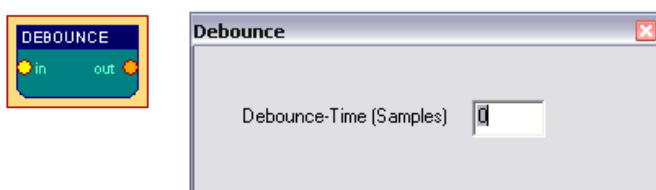
This element calculates the cross-correlation between the two signals connected to input 1 and input 2 in a selectable *Interval* of samples. Correlation is a measurement for the linear relationship between the two signals. The output will be between 1 (fully correlated) 0 (uncorrelated) and -1 (fully inverse correlated).

5.2.4 Counter / Display



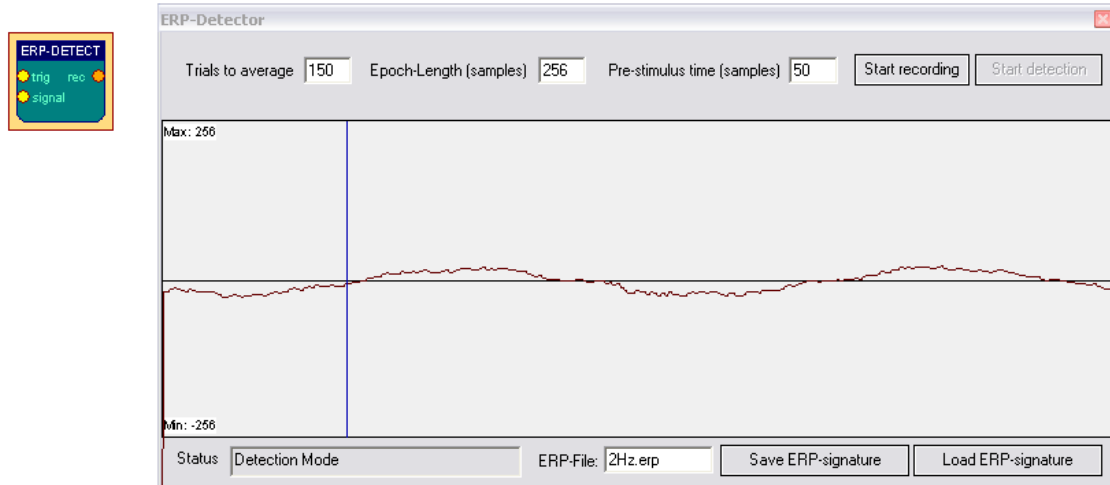
Depending on the selected type, the *Counter* element is used to count transitions from or to INVALID_VALUE, measure frequency of true-false transistions (in Hz) or show the plain input value without counting. The value of the counter is presented at it's output port, and, if desired, in a seperate window as an integer or float value. The counter can be set to an initial value by pressing *Reset Counter Value*. This reset of the value will also be done when the session is resetted using the status bar. Colors of text and background and font size can be selected in the *Display*- section.

5.2.5 Debounce



The Debounce element can suppress frequent changes to `INVALID_VALUE`. This could be used to constrict the generation of events when a signal is close-by a threshold level. The interval is given in samples.

5.2.6 ERP - Detector (Pattern Recognition)

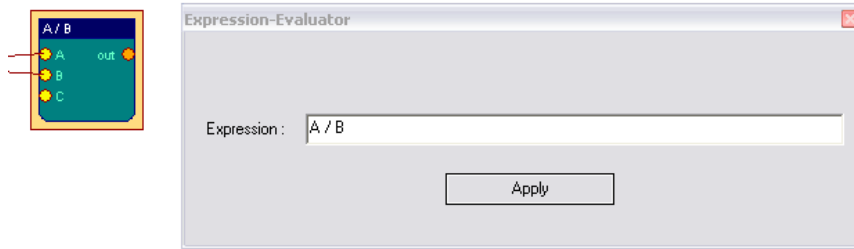


The *ERP-Detect* element can record and detect signal patterns in time domain. It can be used to perform trial averaging, what is a popular technique for extracting event related potentials (ERPs) from a noisy EEG signal. The *Epoch-length* of the trials, the pre-trigger interval and the number of trials can be written into the input fields. The display range is obtained from the *signal* input port. Adjust this range by right-clicking the input port.

The button *Start recording* activates the recording- and averaging-mode of the element. In this mode, the element waits for a TRUE value on it's trigger input port. This trigger signal could come from a threshold - element, from an external switch-button or from a signal generator etc. After the trigger signal has been received, *epoch-length* samples are recorded and added to the internal buffer of the element. Then the next trigger input is awaited and the next trial is recorded. After a number of trials, the noise gets less due to the averaging process and the extracted signal remains. This signal can be saved to an .erp file.

When the *ERP-Detect* element is not in it's recording state, it's continuously calculates the linear difference between the recorded signal and the last *epoch length* samples that came into it's *signal* input-port. The output-port presents the similarity with the recorded signal in percent (0-100).

5.2.7 Expression Evaluator



The *Expression Evaluator* can have up to six input ports, which automatically extend when a signal is connected. The input signals are referred to as A, B, C, D, E, F and can be combined with decimal constants, elementary functions, unary and binary operations to a mathematical expression that is evaluated when the session is running. The result of the expression is presented at the element's output port. The evaluator implementation builds upon the open source library GNU-Libmatheval. (see <http://www.gnu.org/software/libmatheval>)

Supported elementary functions are:

- exponential (exp), logarithmic (log), square root (sqrt), sine (sin), cosine (cos),
- tangent (tan), cotangent (cot), secant (sec), cosecant (csc),
- inverse sine (asin), inverse cosine (acos), inverse tangent (atan), inverse cotangent (acot), inverse secant (asec), inverse cosecant (acsc),
- hyperbolic sine (sinh), cosine (cosh), hyperbolic tangent (tanh), hyperbolic cotangent (coth), hyperbolic secant (sech), hyperbolic cosecant (csch), hyperbolic inverse sine (asinh), hyperbolic inverse cosine (acosh), hyperbolic inverse tangent (atanh), hyperbolic inverse cotangent (acoth), hyperbolic inverse secant (asech), hyperbolic inverse cosecant (acsch),
- absolute value (abs), Heaviside step function (step) with value 1 defined
- supported unary operation is unary minus ('-').
- supported binary operations are:
 - addition ('+'), subtraction ('-'), multiplication ('*'),
 - division multiplication ('/') and exponentiation ('^').

Usual mathematical rules regarding operation precedence apply.

Parenthesis '(' and ')') could be used to change priority order.

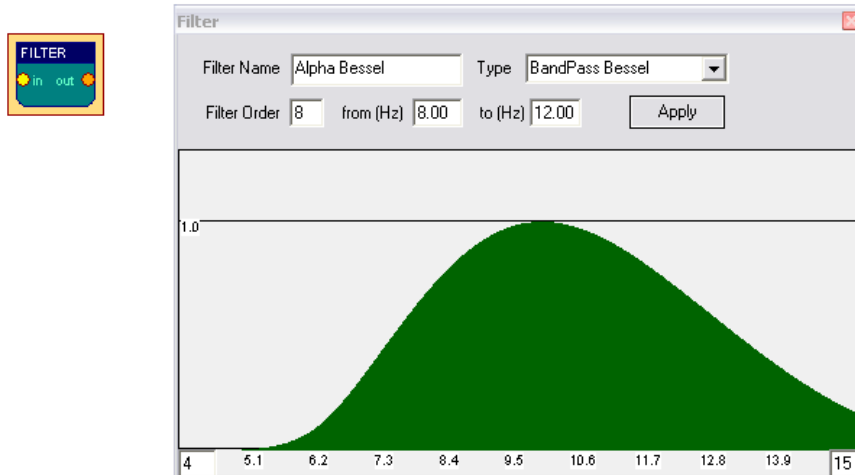
Blanks and tab characters are allowed in string representing function; newline characters must not appear in this string.

5.2.8 Filter

The Filter element provides digital low-pass, high-pass, band-pass or band-stop filters. A low pass filter will filter out high frequencies and let low frequencies pass the filter. A high pass filter does the opposite - it will filter out low frequencies and let high frequencies pass. A band pass filter will filter out lower and higher frequencies and let middle frequencies pass the filter. A band stop filter will filter out the middle frequencies and let lower and higher frequencies pass the filter.

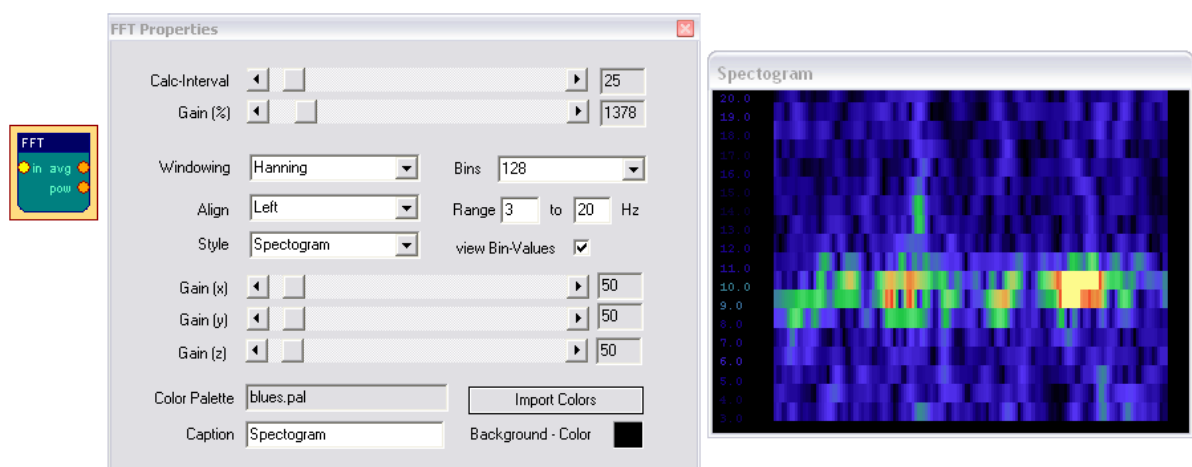
The filter can have *bessel-* or *butterworth -* type. The *bessel*-filter has a slower roll-off in frequency domain (meaning it is less precise in attenuating around it's corner frequencies) but does not have so much ringing (overshooting) in time domain.

The *Filter Type*, *Filter Order* and the frequency - limits for the filter can be selected in the user dialog. A higher filter order gives a sharper response in frequency domain but a longer delay in time domain. When the button *Apply* is pressed, the filter response in frequency domain is shown in the given display range. The display range can be adjusted by typing new values in the bottom left and bottom right fields.



The filter shown above lets frequencies around 10Hz pass without attenuation. Frequencies below 6 Hz or above 14 Hz are attenuated heavily. For more information on the usage of filters see the design examples. The *Filter* - element uses the free Fid-Lib library by Jim Peters and Tony Fisher (see <http://uazu.net/fiview>)

5.2.9 FFT (spectral analyser)



The *FFT* element performs a fast fourier transformation on the connected signal. This shows the frequency components that are present in the signal, just like the ear can

detect the different tone-heights that are present in a sound. Using the FFT -element, the changing frequency-components of for example brainwaves can be displayed as a bar-graph, spectrogram or 3d- landscape. The kind of display can be changed using the *Style* combo-box.

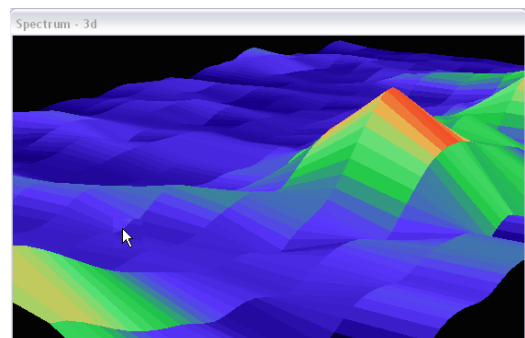
The *Calc-Interval* sets the number of samples that are buffered before the next transformation is done. An interval of 25 samples gives about 10 transformations per second at 256 Hz sampling rate.

The number of *Bins* defines the distinct frequency components (= bands). Many bins give good frequency resolution but slow reaction in time. *Range* selects the bands to be displayed (in Hz). *Align* (left, right, bottom) rotates the display.

A color palette can be used to map the intensity information to colors, what is necessary in the 2d and 3d - views (for usage of the color palettes see palette editor).

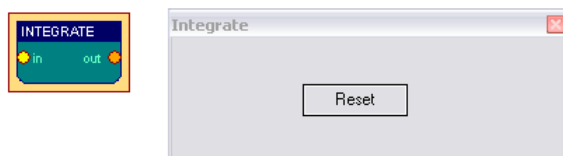
The *Gain (%)* amplifies or attenuates the signal before the FFT is done. The Gain(x), -(y) and -(z) affect the display and are effective in the 2d and 3d views.

When using the 3d -display, left- or right-clicking and dragging the display window will change angle or position of the view (see section design examples).



The FFT-element has two output ports: *Average* and *Power*. The first gives the average frequency of the bands in the selected range, the second gives the average power (magnitude) of the bands in the selected range.

5.2.10 Integrator



The *Integrator* element continuously sums the incoming values. Pressing *Reset* sets the buffer to zero, this is also done when starting or resetting the session.

5.2.11 And, Or and Not



The logical *And*, *Or* and *Not* elements perform logical operations on the input values and output the result. BrainBay has a special signal value which is treated as “False”.

This signal value (-32767) is called “INVALID_VALUE” and is created e.g. by the *Threshold* element when the incoming signal exceeds the positive or negative thresholds. The logical elements *And*, *Or* and *Not* take INVALID_VALUE at their input ports as logical “False” and all other values as logical “True”. They behave like standard logical operations:

And outputs “True” when none of its inputs is INVALID_VALUE
Or outputs “True” when at most one input is INVALID_VALUE
Not outputs “True” when the input is INVALID_VALUE

(else the elements output “False”).

The logical elements provide some flexibility how the results “True” or “False” are represented at their output port: Using the provided comboboxes in the property window, the behaviour for “True” and “False” can be selected separately which increase the flexibility and usage applications of these elements:

The output behaviour for “True” can be set to:

- value 512
- the value of input port 1
- the maximum of input 1 and input 2
- an arbitrary numeric value (which can be defined in an input field).

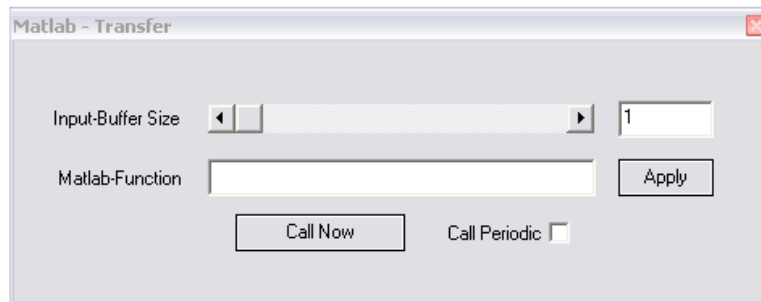
The output behaviour for “False” can be set to:

- INVALID_VALUE (-32767)
- an arbitrary numeric value (which can be defined in an input field)
- no output at all



Alternatively, a “binary” operation mode can be selected for the logical elements. If this mode is selected it overrides the output behaviour selected in the above comboboxes. The binary operations perform a bitwise and (&), or (|) or xor (^) operation with the values of input1 and input2 (or with the given numeric value in case of the xor operation) – similar to the bitwise logical operation in the C-programming language.

5.2.12 Matlab Transfer



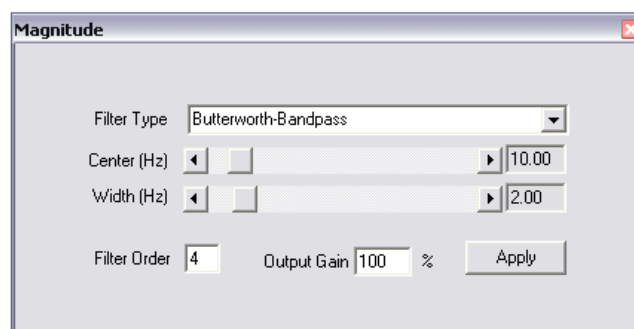
The Matlab - element can be used to transfer sample buffers to the Matlab application for further processing (see <http://www.mathworks.com>). Matlab is commonly used in academic and scientific signal processing. As this is a commercial application, the needed .dlls cannot be delivered with brainbay, and the Matlab-element will only work if you have a licenced version of matlab on your system.

The element can process up to six input ports named A, B, C, ...

During operation, the input values will be collected in buffers of given size. By pressing Call Now, the buffers are transferred to Matlab and can be accessed via the Matlab-variables A, B, C ... and a Matlab-function of given name will be called. The result (the ANS-variable) will be transferred back to BrainBay and presented on the output of the *Matlab*-element.

When *Call Periodic* is selected, the transfer will be done automatically when a new sample arrives, this will only work for matlab-functions with low complexity / short execution time. (Please note that this element is probably outdated, it's correct operation needs to be verified with current versions of Matlab).

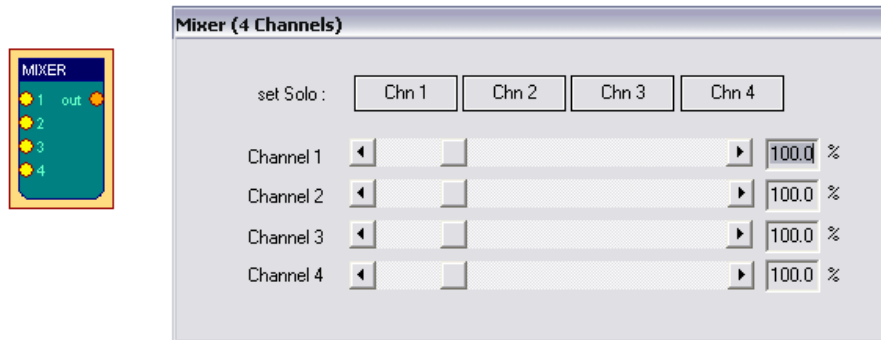
5.2.13 Magnitude



The *Magnitude* - element can be used to calculate the activity in a certain frequency - range of a signal. This is often referred to as 'power in pass-band'. The filter type that is selected via a combobox can be *Bessel-Bandpass* or *Butterworth-Bandpass*. (for more details on filters see the *Filter* - element or Jim Peter's *FiView* - application, <http://uazu.net/fiview>). The *Center (Hz)* and *Half Width (Hz)* - parameters adjust the pass-band, where activity shall be measured.

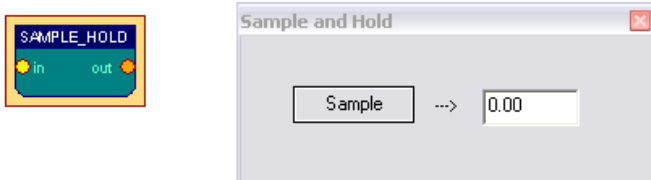
An Example: when you want to measure Alpha-frequencies of brainwaves in the range from 10 to 12 Hz, select a center of 11 Hz and a half width of 1 Hz for the filter. The filter order controls the 'sharpness' of the cut-off of unwanted frequency-ranges. A higher order gives better cut-off, but the response of the filter in time will be less accurate. The *Magnitude* - element outputs the activity in the given passband, a gain value (in %) can be applied to amplify the output.

5.2.14 Mixer



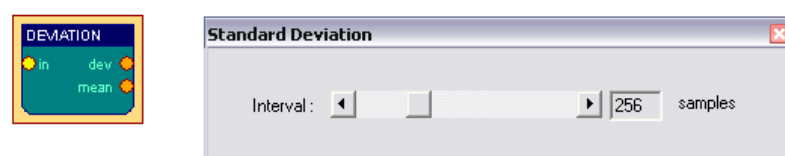
The *Mixer* - element can be used to merge up to four input signals into one output signal. The amplification-ratios are given with the corresponding slider-bars. Using the buttons Chn1 - Chn4, presets for solo-output of the selected channel are activated.

5.2.15 Sample and Hold



The *Sample-Hold* - element stores the current input-value when the button *Sample* is pressed. This value will be present at the output-port until another value is stored using the button.

5.2.16 Standard Deviation



This Element calculates standard deviation and mean of n samples and puts the results to the output ports. The number of samples for the calculation interval can be set using the slider bar.

5.2.17 Threshold / Meter

The Threshold - element combines following useful functionalities:

- Averaging and amplification or attenuation of the incoming signal (using the slider bars in the *Preparation*-section)
- Setup of fixed upper- und lower limits for the signal (thresholds)
- Generation of dynamic upper and lower limits, using a quantile, percentage of min/max or percentage of the averaged values of the previous n samples
- Calculating a baseline value for n seconds after the session starts
- Passing only rising or falling values
- Defining the output behaviour for valid or invalid values; the default behaviour is that if incoming values are not in the specified range, the element outputs INVALID_VALUE, otherwise the signal will be passed to the output port
- Display of a *Meter* - window to show the current value of the signal and the upper and lower bounds, which allows manual adjustment of the threshold value via the *cursor-up* / *cursor-down* keys



The minimum and maximum values of the upper/lower limit sliders are obtained from the input port signal range. This range can be adjusted by right-clicking the input port.

When dynamic threshold calculation is used, the upper and/or lower limits are calculated periodically from the last *interval length* number of samples. The calculation method can be selected from the combo boxes (quantile, % of min/max or % of mean value).

By enabling *calculate only once (get baseline)*, the dynamic thresholds will only be calculated at the beginning of the session (please note that the interval for the calculation of the baseline is then given in seconds - not in samples. E.g. an interval of 20 will average 20 seconds of incoming values at the beginning of the session to determine the threshold).

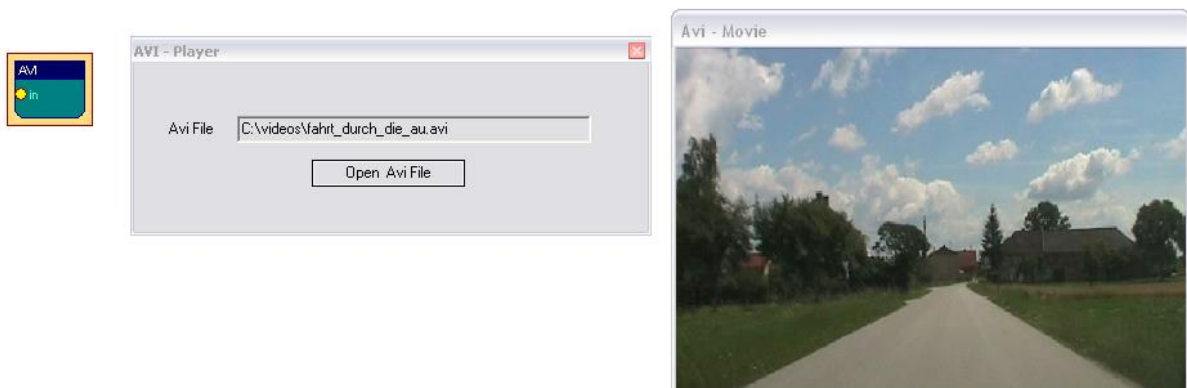
If *rising-* or *falling values* is selected, only progressive or regressive values will pass the threshold. In the *Meter Window* - section, font size, bar size, colors and caption for the meter display can be selected.

The *Meter Window* section allows various color and style adjustments of the meter window. If desired, the meter window could also be hidden. **When the meter window is displayed and has the input focus during a running session, users can adjust the given upper threshold by pressing the *cursor-up* or *cursor-down* keys on the keyboard** – thus, the threshold level can be easily adjusted on-the-fly.

The Threshold element is a useful tool for feedback-purposes, where the reaching a certain state (high activity or low activity) is desired. Combined with a *Midi* - element, feedback tones can be generated if a desired level is reached.

5.3 Signal Targets

5.3.1 AVI – Player



The AVI-Player element can be used to display and navigate avi-movies. The value that comes into the input-port of the element is interpreted as frame number. This frame will be shown in a separate display window. Using rising or falling values, the movie can be watched forward or backwards. The sound of the AVI-movie will not be present at playback (use the *Media Player* element to include sound output). A video-codec that can decode the given avi-file has to be installed on the system to use the AVI-Player element. Please note that the *MediaPlayer* element is also available which provides other / additional features and supports different media file formats.

5.3.2 Com- Writer



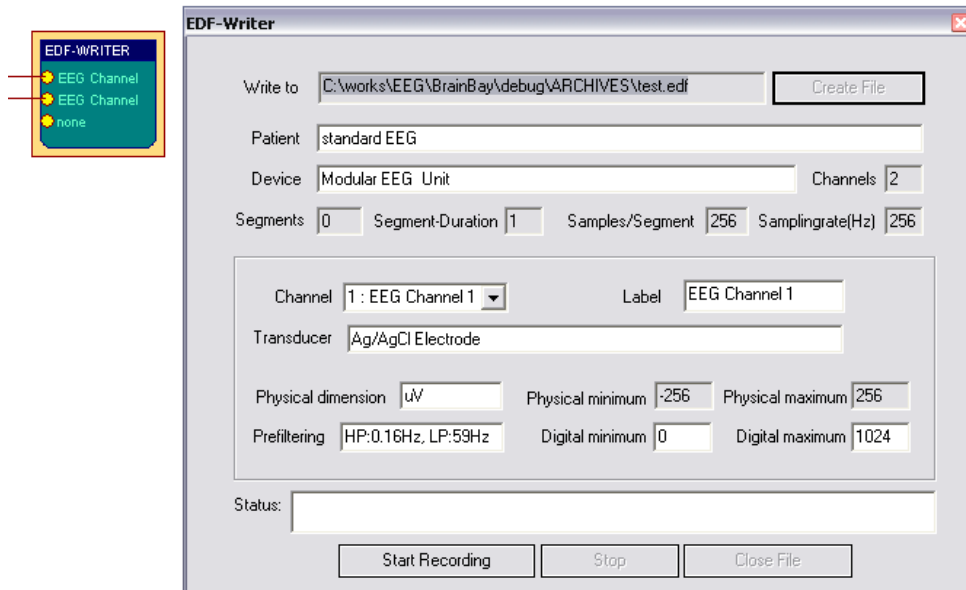
The *COM-Writer* element can be used to send command and data values to a connected Monolith-EEG amplifier in bidirectional mode (using P21 firmware protocol) . Using the Com-Writer, runtime options of the MonolithEEG like baud- or sampling rate can be changed and the digital I/O- ports of the MonolithEEG can be set. One command/data frame consist of three bytes that can be defined using the input fields of the user dialog. The frame is sent by pressing the button. When *triggered sending* is enabled by the user option, the frame will be sent every time an input different from INVALID_VALUE is received by the *trigger* input-port.

5.3.3 EDF- Writer

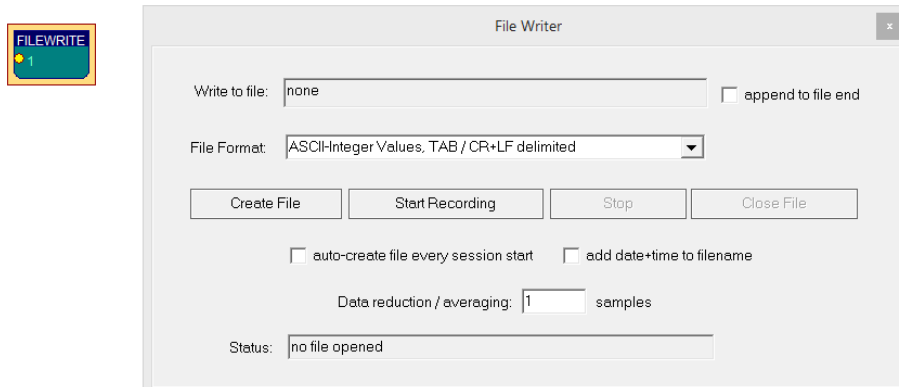
The *EDF-File Writer* element can be used to created a biosignal archive in European Data Format (see <http://www.edfplus.info>). The header of an EDF file stores various information about the recording: date, patient data, recording device, sampling rate, data segments, signal ranges and descriptions for the data channels. This information can be set using the data fields of the user dialog.

The description for the connected channels can be set using the *Channel* combo-box and the corresponding data fields. When all channels have been connected and labelled, the button *Create File* can be used to choose location and file name for the EDF file to be created.

The recording of channel data can be controlled by the buttons *Start Recording* and *Stop Recording*. After recording, don't forget to press *Close File* to finish the writing process and close the file.



5.3.4 File – Write



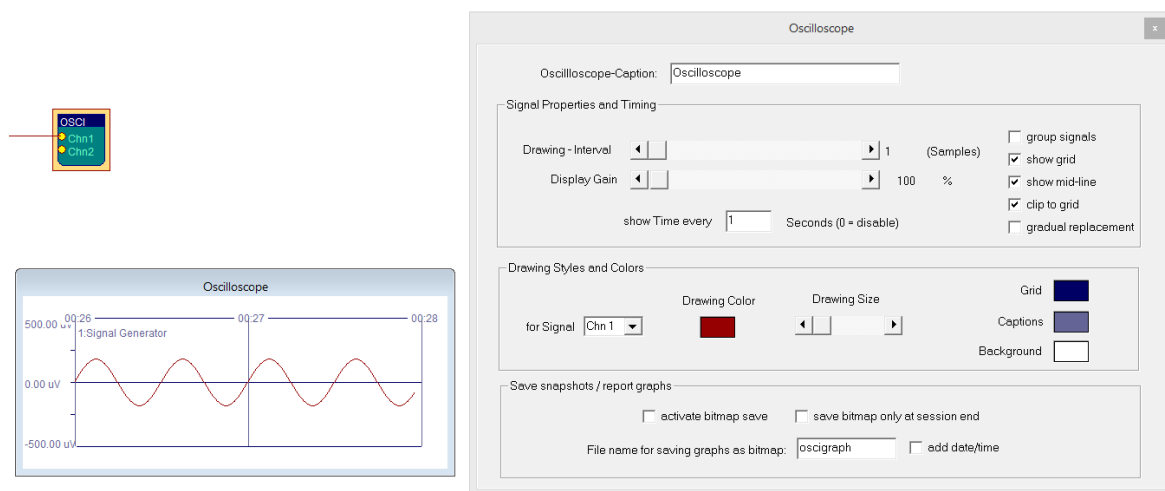
The *File-Write* element can be used to generate an archive file with channel values, which allows processing of the data in other applications. The *File-Read* element can be used to read the channel values from this archive file – usually this will be done in another design configuration.

Using the *File Format* combo box, the type of storage and the delimiters for rows and columns can be selected. When using *ASCII-Integer Values*, a human-readable text file will be generated. If more than one channel is connected to the element, the channel values will be written as columns with commas (CSV- text file) or TABs as column separators. As delimiters for the rows, CR/LF - delimiters can be generated. These text files could be imported in Microsoft Excel or other applications for display and further processing. Further available file formats are *Bioexplorer File Format* (including a header which makes this file compatible to the Bioexplorer application) or *1-channel 16 bit raw* format, a one-channel binary file that could be imported in a sound-processing application which supports .wav format (note that in this case only one channel will be processed).

The archive file can be created and closed manually via the dedicated buttons in the configuration dialog. Alternatively, the File-Writer element offers the feature *auto-create file every session start*: if enabled, the archive file will be automatically created at session start and closed at session end. To prevent overwriting previous archives, the option *add date+time to filename* can be used (so that unique filenames containing the creation date/time will be created).

If desired, a number of samples can be averaged before stored to the archive file – for example when a value of 1000 is entered in the field *Data reduction / averaging* only one (averaged) value will be written to the file every 1000 samples.

5.3.5 Oscilloscope



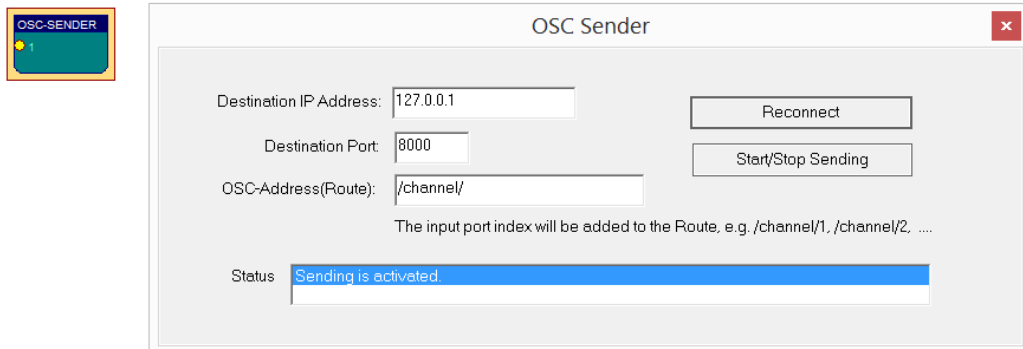
The *Oscilloscope* element is essential for the display and inspection of one or multiple signals. The connected signals are shown in a separate window which can be freely resized. During runtime, a data grid showing the signal ranges and the current time can be displayed. The *Drawing-Interval* slider sets the display granularity for the oscilloscope: a selection of 1 will display every incoming value, a selection of 4 will display every fourth value etc. The *Display-Gain* slider is used to amplify / attenuate all connected signals. **The gain and drawing-interval settings can also be changed during session runtime by using the cursor keys if the oscilloscope window is selected (up/down for gain, left/right for interval).**

If the *group* option is used, the connected signals will be shown in the same data grid (in this case, the signal range of the first connected channel is used for all connected signals). The settings *show grid*, *show mid line*, *clip to grid* and *gradual replacement* determine if a grid is used, when the vertical bars for time periods are inserted, how the signal traces are drawn and if they are allowed to exceed their dedicated charts. In section *Drawing styles and colors*, background color and individual signal colors (per channel) are adjusted using the color-picker-buttons and the signal trace width can be adjusted.

The section *Save snapshot / report graphs* provides allows storing the oscilloscope traces before they get removed: if *activate bitmap save* is enabled, a snapshot is taken of the oscilloscope window and saved under the given filename in the subfolder */Reports* (.bmp will be added automatically). Optionally, date and time can be added to the filename to avoid overwriting the bitmap file.

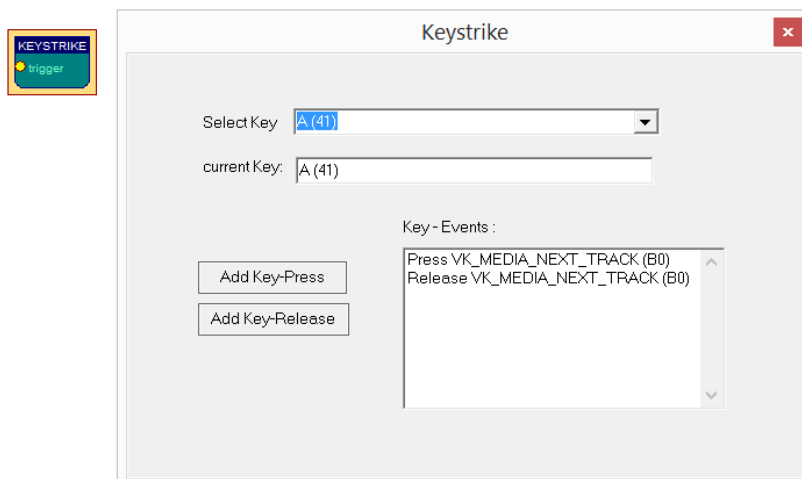
If the option *save bitmap only at session end* is active, the snapshot will only be taken when the session end is signalled by the *Sessiontime* element. See description of the *SessionManager* and *Sessiontime* elements for how to use report graphs of oscilloscopes to show training progress.

5.3.1 OSC-Sender



The *OSC-Sender* element can be used to transfer live data from Brainbay to other applications via UDP / Open Sound Control (OSC) protocol. Only a UDP server listening on the Destination IP-address / port is necessary to receive the information. Multiple signals can be connected to the *OSC-Sender* element. Every connected signal is transferred via a dedicated OSC route (address), which is composed of a selectable prefix (eg. /channel/) and the appended channel number. Please note that every OSC route identifier must start with a slash. For details about the OSC protocol structure see: http://opensoundcontrol.org/spec-1_0-examples

5.3.2 Keystrike



The *Keystrike* element allows the generation of virtual keyboard input (i.e. key press and key release events which behave like real keyboard input in most applications). The key actions can be added to a listbox. During session runtime, the key events are generated when a connected signal changes from "INVALID_VALUE" to a valid value (e.g. when a threshold value in a preceding threshold element is reached).

5.3.3 Particle Animation



The *Particle Animation* element provides a graphical particle system that can be controlled using signal values from up to six input ports. The particle system can produce nice visual effects like color fountains, flames, star-flights or, in combination with signal generators, moving circles and other geometric figures in 3d. The System can be influenced by changing the current parameter using the *Range* slider-bar.

Parameters for the Particle System:

- *Number of Particles, Generation Interval*
- *Slowdown, Color*
- *X-, Y-, Z- Position*
- *X-, Y-, Z- Speed*
- *X-, Y-, Z- Gravity*
- *Life Span, Randomizer*

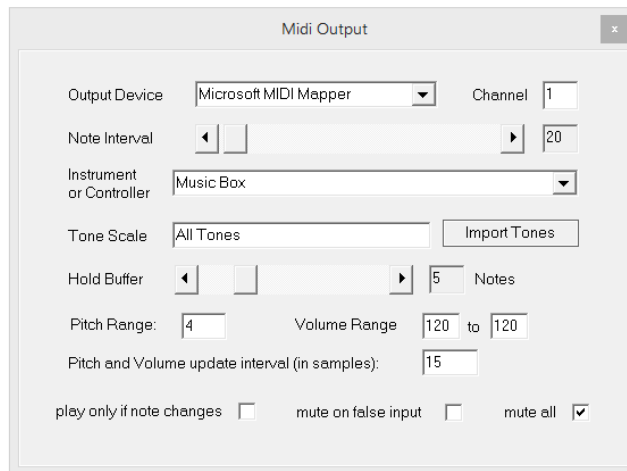
Recommended usage of the Particle System:

- select a color palette for the particles (see palette editor)
- choose one of the above parameters from the combo box and adjust it's value to gain a desired effect
- repeat previous step for other parameters, until you have the effect you like
- connect a signal to input port 1 of the element
- select a parameter that suits well to the type of feedback you want
- select *Remote Control by 1*, now the value of this parameter will change according to the signal connected to input port 1.

Use the *get minimum* and *get maximum* buttons to set the current input-value as new minimum or maximum for the parameter value. This way, the upper and lower bounds for remote-controlling the parameter can be set.

When *reset min and max* is pressed, the min and max values will be reset to the original values for this parameter. The best results for the particle system can be gained by using an OpenGL-accelerated graphics card.

5.3.4 Midi Output



The *Midi* element provides Midi-Audio feedback for the connected signal. Toneheight and volume are controlled by the connected input ports. The signal range of the connected input signals is mapped to the selected volume range or tone scale (note that the range can be modified by right-clicking the input ports).

The Output Interval sets the frequency of tone generation: an interval of 1 will generate a midi tone every time a new value arrives (this is usually much too fast). A setting of 25 gives about 10 tones per second when a sampling rate of 256 is used, which is still very accurate. The interval can be directly set via the dedicated input port as well.

The *pitch* input port allows adjusting the pitchwheel setting according to the incoming value and its range. The maximum span for pitch and volume can be defined via the settings *Pitch Range* and *Volume Range* respectively.

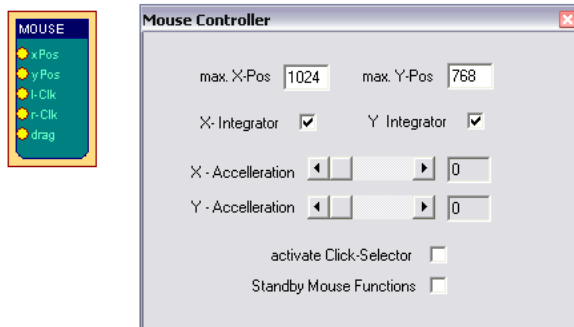
For *Output Device*, a desired Midi-Output device can be selected (please note that this device must be enabled in the Options Menu before). Using Channel 1 - 16 allows multiple midi-elements with different instruments to work in parallel (while channel 10 usually selects the Drum-Set).

The combo-box *Instrument or Controller* selects the midi instrument or a midi-controller from 0 - 127. Using midi-controllers makes sense if an external midi-device like a synthesizer or drum-computer is connected, or a Midi-software-router to transfer the midi-commands into other running sound applications is available (for example LoopBe1 freeware bridge).

With the button *Import Tones*, a tone-scale which has been generated with the *Tone Editor* can be imported and used as a harmonic filter for the midi-notes. (see *Tone Editor*).

The *Hold buffer* sets the number of tones that will be left *on* at the same time. When *play only changes* is selected, two consecutive notes will only be played if they are different.

5.3.5 Mouse Controller



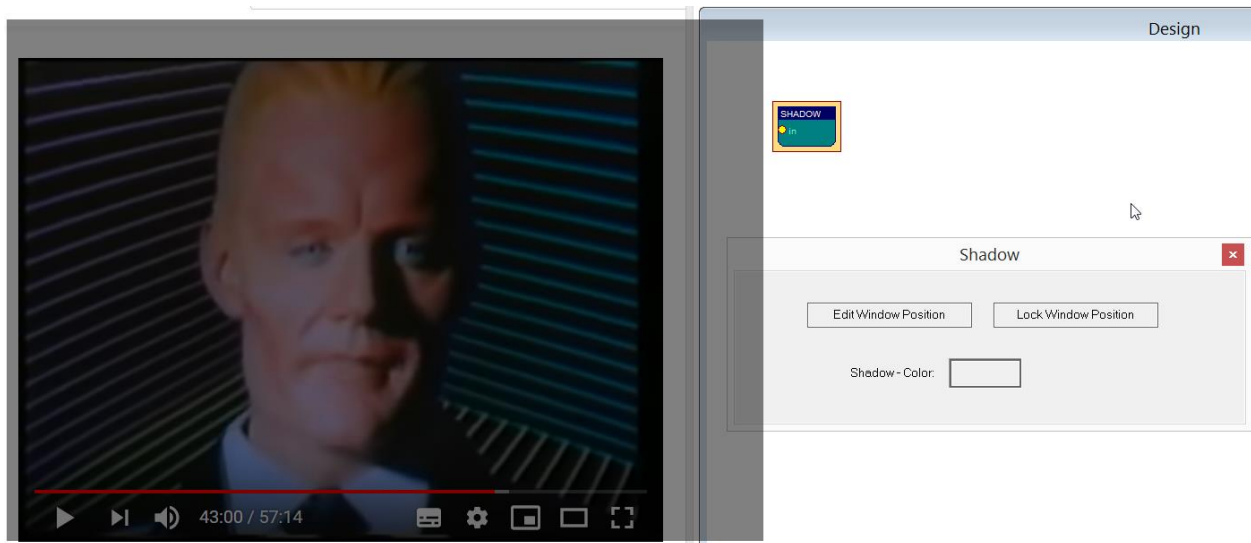
The *Mouse Controller* element provides an interface to the mouse-cursor and clicking functionalities. The current screen-resolution should be set as maximum X- and Y-positions for the cursor. During session runtime, the present values at the *xPos*- and *yPos*- input ports will influence the cursor position. If the *x-Integrator* and *y-Integrator* options are selected, the input values are interpreted as relative changes (Joystick mode), if not, absolute positions will be used.

A left click will be performed when a value other than `INVALID_VALUE` is passed to the *l-Clk* input port. Right-clicks and drag-clicks can be performed in the same way, using the *r-Clk* and *drag* input ports.

The option *activate Click-Selector* displays a separate tool-window which allows selection of right-, double- or drag-clicks without using the corresponding input port. Thus, a right-, double- or drag-click can be performed by choosing the type of the next click in the click-selector window, and the special click can be performed with the normal left-click action. The *Dwell-Time* can be used to perform a left-click operation by holding the mouse cursor in a given screen-area for a given time. Thus, clicking can be performed without having a separate control signal connected to the *l-Clk* port.

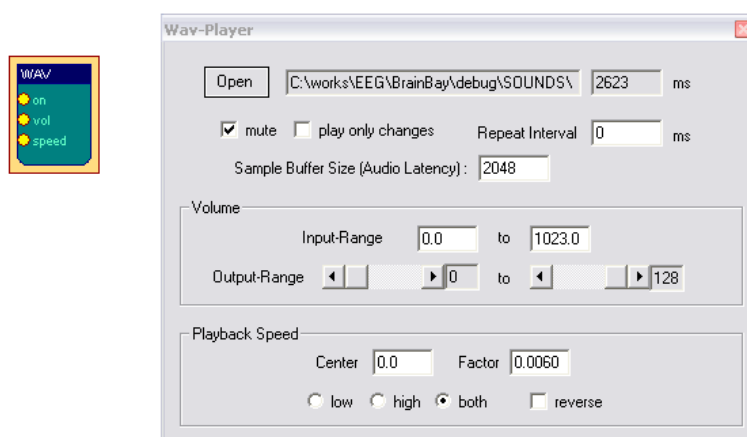
A useful source for the *Mouse Controller* could be the *Camera* element (feature tracking enabled). After some filtering and scaling, the estimated face positions that are put out by the Camera element set the cursor position for the mouse and give a functional head-mouse interface (see design examples).

5.3.1 Shadow



The *Shadow* element displays a semi-transparent window with selectable dimension and color above other windows on the desktop. The transparency / occlusion can be controlled via the input port. This can be used for feedback/reward purposes – e.g. to show a picture or movie which is displayed by another application. The shadow window is placed on top of the respective content. The input port range (right click on input port) is mapped to the transparency of the shadow window, i.e. if the minimum value is sent to the input port, the window is fully opaque, and if the maximum value is sent to the port, the window is fully transparent. Please note that the window can only be moved / scaled if the edit mode is enabled in the properties, and that the transparency only works if the window is locked. So the recommended procedure is to place the window over a movie and then lock the window (and eventually save the design).

5.3.2 Sound Player



The *Sound-Player* element can be used for triggered playback of a short audio-file. Supported file formats include wav, mp3, voc and aiff. The file could contain sound effects or reward tones of about 1 to 20 seconds length. The element is not suited for longer music files because of long loading / conversion times (use the *Media Player* element for longer files). The input-ports control starting of the sound (on), the current

volume (vol) and the playback speed / toneheight (speed). The sound will be played when the *on* input is different from INVALID_VALUE. If the option *play only changes* is selected, the sound will be played only once if the value of the *on* input-port does not change. The *Repeat Interval* can be used to select a minimum duration until the sound is triggered next time, thus a pause can be introduced.

The *Sample Buffer Size* sets the size for the internal playback buffer, this value affects the audio latency / system performance. Using values < 4096 is recommended for accurate playback.

The ranges for volume (*Input-Range*) and speed (*Center* and *Factor*) are obtained from the connected input signals and can be set manually by changing the values in the user dialog. The *reverse* option changes speedup or slowdown of the sound when the *speed* input values rise. The *Center* - value sets the value for playback at original speed and the *Factor* - value sets the gain for speed-changes.

5.3.3 TCP – Sender

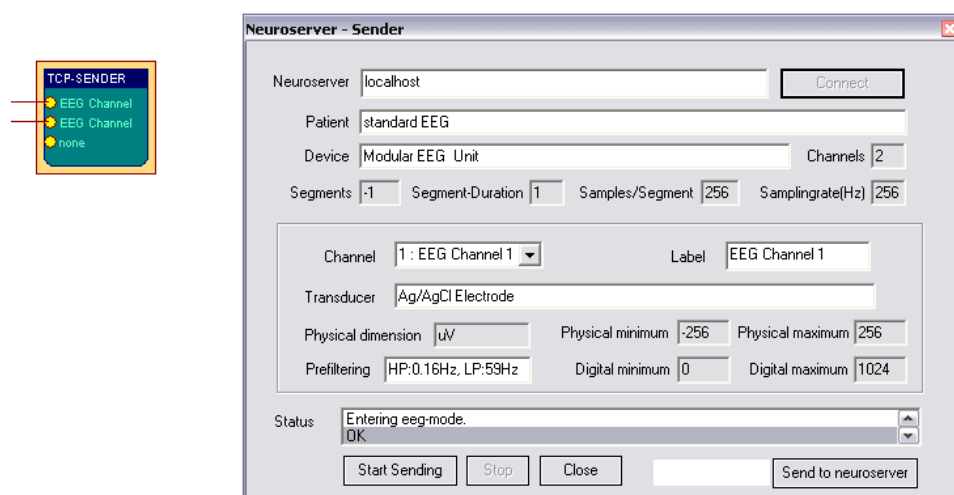
The *TCP-Sender* element can be used to transfer signal channels via network, using the neuroserver software by Rudi Cilibrasi. To establish connection, a neuroserver has to be running on a known host computer in the network or on the local host. (see <http://openeeg.sourceforge.net/doc/sw/NeuroServer>).

The *TCP-Sender* element then connects to this service as a client which sends bi-signal data in EDF-format.

Connecting the TCP-Sender element to the neuroserver service

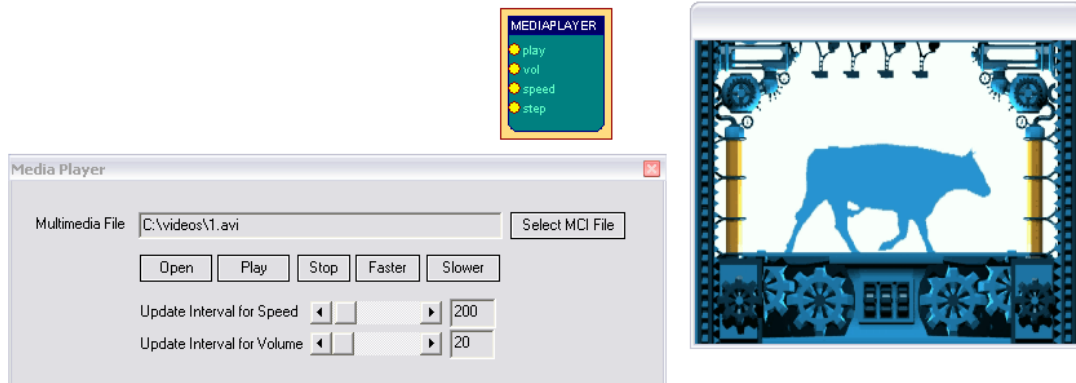
As in the EDF-Writer element, the first thing to do is to fill out the descriptions for the EDF data header and channel information (see EDF-Writer element). The signal range for a specific channel is obtained from the connected input signal, it can be modified by right-clicking the input port.

To connect to the neuroserver, enter the IP-adress of the host computer where the service is running into the field *Neuroserver* of the element's user dialog. (when running a local service type 'localhost'). Press the *Connect* - button. In case of a successful connection, the *Status*-listbox will show the lines 'Entering EEG mode' and 'OK'.



Start Sending and *Stop* are used to control the data flow to the neuroserver. The button *Send to neuroserver* directly sends a neuroserver command written in the text field to a connected service. Use *Close* to disconnect from the service.

5.3.4 Media Player



The *Media Player* element uses the Windows MCI interface to playback video or sound-files. AVI, WMV, WAV, MP3 and other formats could be used. After a media-file has been selected, it is played by receiving a value different from `INVALID_Value` at its *play* input port. Volume and playback speed can be adjusted by the *Vol* and *speed* ports, where a value of 1000 sets full value or normal speed. Update Intervals for *vol* and *speed* have to be set according to system performance because these functions are time-consuming. Inputs into the *step* port cause a video file to step one frame forward.

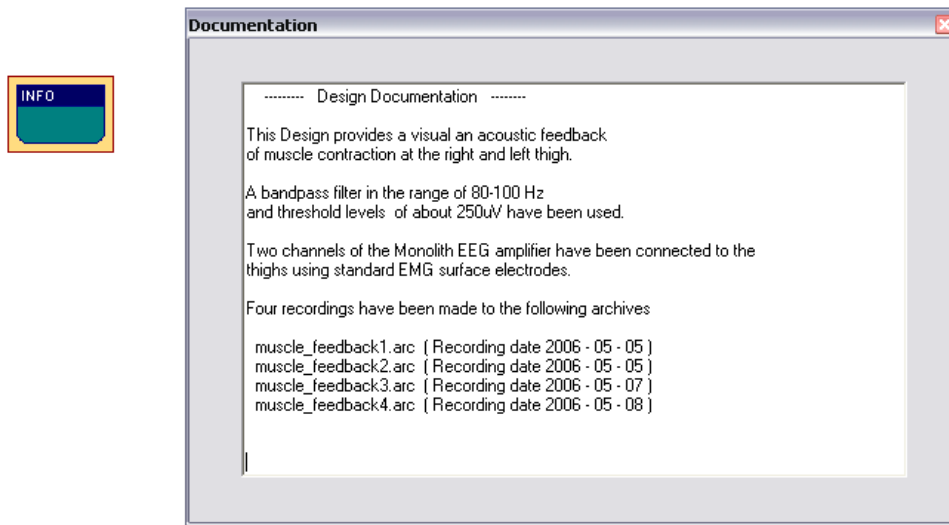
5.3.1 Volume



The *Volume* element can adjust the master sound volume. This can be used to influence the sound output of other applications running in parallel. The input port range (right click on input port) is mapped to the volume, i.e. if the minimum value is sent to the input port, the volume is set to 0% and if the maximum value is sent to the port, the volume is set to 100%. The *Volume* element can be combined with the *Shadow* element to control both audibility and visibility of a movie playback or computer game.

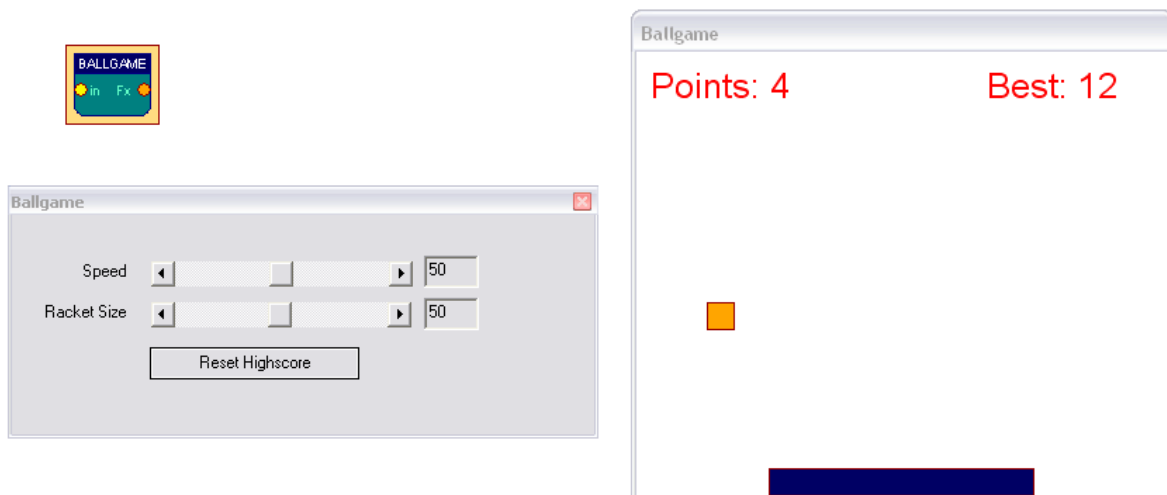
5.4 Other Elements

5.4.1 Documentation



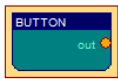
The *Documentation* element simply provides a text box. It can be useful to describe the overall purpose of the design, special design elements or archive files that have been recorded with this design.

5.4.2 Ballgame



The ballgame is an example for a feedback-game, implementing a low-end version of the classic 'arkanoid' - arcade game originating in the early 80ties. The purpose of the game is to catch the ball with the moving bar (the racket). The position of the racket is controlled by the input port of the element. *Racket Size* and *Speed* of the ball can be controlled via the user dialog. The best number of hits is displayed in the game window and stored by the element.

5.4.3 Button



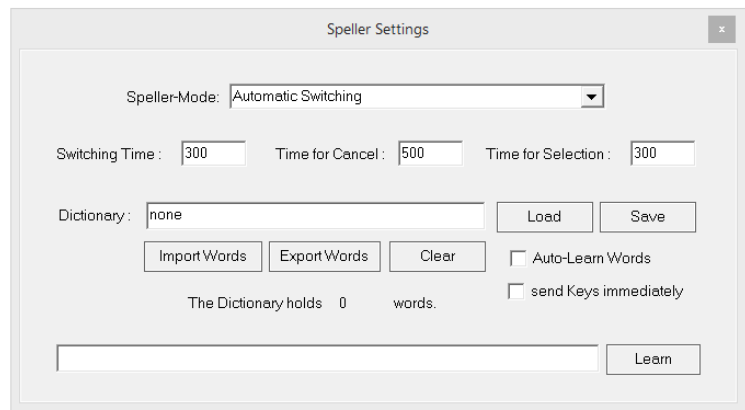
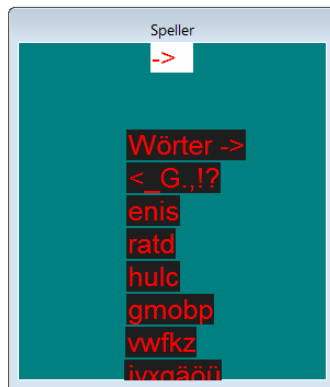
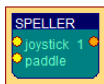
The *Button* element allows placement of a desired icon or symbol inside the main application window, and to select a function which shall be executed when this icon is clicked during a running session. The default file path is set to the */GRAPHICS* sub-folder. Alternatively, a relative filepath to other locations can be given. The file type of the graphics file must be *Bitmap* (.bmp, 24bit color). The functions for the button can be selected via a dropdown combo box. Supported functions for the button are:

- *Start Session, Stop Session, End Session*
- *Send Value1 if pressed else Value2*
- *Send Value1 if pressed else INVALID_VALUE*
- *Toggle Value1 and Value2*
- *Send Value2 for one second, else Value1*
- *Display Device Settings* (only available if Amplifier present in the design)
- *Display Application Settings*

If desired, a window caption can be displayed. If no window caption is displayed and the session is locked (see Application Settings), the button bitmap will displayed completely without frame or border.

5.4.4 Speller

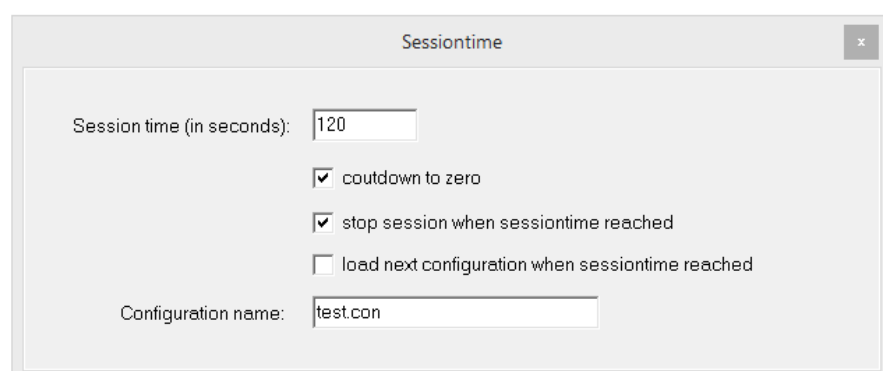
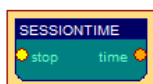
The *Speller* element provides an experimental user interface for step-by-step selection of letters (e.g. via EMG muscle activity, eye blinks, IMU-accelerometer values etc.) for the purpose of writing and/or using a speech synthesizer.



The GUI of the element shows rows with a number of letter- or word-suggestions. The rows can be automatically switched / highlighted or browsed manually (via the paddle input). When a row is selected via the incoming values of the joystick/paddle input ports, the individual letters (or words) can be chosen in a subgrid. A dictionary of words can be specified which allows word-prediction for more efficient use. If the option *send Keys immediately* is enabled, keystrokes of written letters or words will be created on the local system.

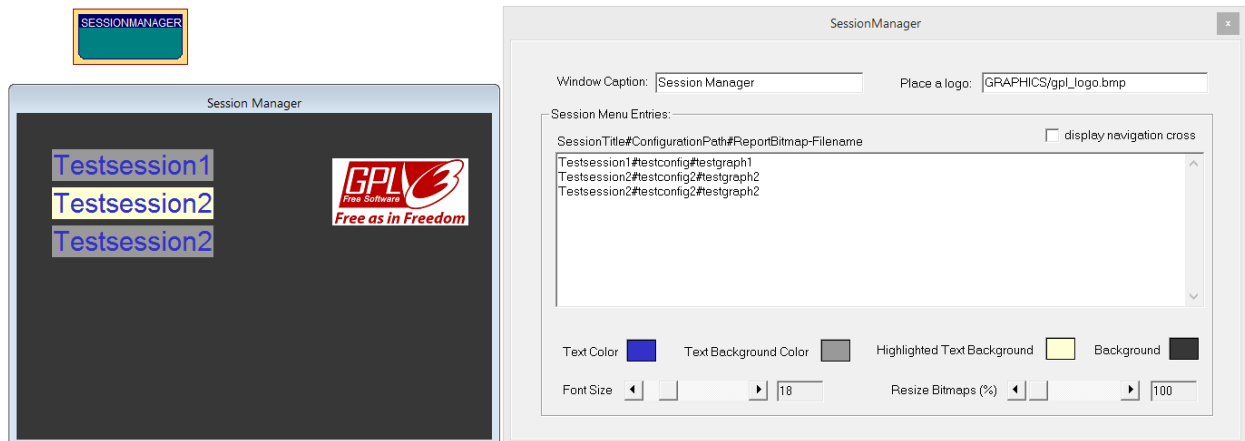
5.4.5 Sessiontime

The *Sessiontime* element allows definition of the session duration in seconds. Depending on the selected options, the session can either be stopped, or a desired configuration design file can be loaded. If the *countdown to zero* option is activated, the element will output the remaining seconds (e.g. in order to display this value using a counter/display element). If the *load next configuration* option is selected, another design configuration will be automatically loaded when the sessiontime is reached. Per default, design configuration files are expected in the *Configurations* subfolder. A possible use case for this parameter is to switch back to a menu which is displayed using the *SessionManager* element.



By sending values to the *stop* input port of the element which are different from *INVALID_VALUE*, a running session can be stopped.

5.4.6 SessionManager



Using the SessionManager element, a number of design configurations can be provided to the user in form of a selection menu. In order to load a desired design, the entry in the list can be selected via keyboard (cursor up/down to change element, press enter to select) or via a dedicated navigation cross (which can also be used on tablets with touchscreen).

Optionally, a collection of oscilloscope report graphs which have been saved in previous sessions can be displayed. The next/previous report graph for the highlighted design can be shown by pressing the cursor keys (left/right) or by using the navigation cross.

The menu entries are composed in a listbox which holds one line per menu entry: Every line consists of 2 or 3 string items separated by a # character. The first string item represents the title of the menu entry which will be displayed in the SessionManager GUI (the session title). The second string item defines the name of the design configuration file which shall be loaded upon selection of the menu entry (given as relative path to the */Configurations* subfolder). The (optional) third string item specifies the name of the report graph files for this design configuration - see description of the oscilloscope element. It is expected that the oscilloscope element which generates the report graphs is configured to automatically add date and time to the file-names (to prevent overwriting previous report graphs). The extension .bmp is automatically added to the given report graph file name, the files are expected in the subfolder */Reports*. If .bmp files are found, they will be displayed in the lower third of the SessionManager GUI.

The menu can be further customized in terms of font size, font color, highlighting color and background color and the report bitmaps can be resized to a given percentage of the original.

On the next page, an example of a customized menu including session report graphs can be seen.

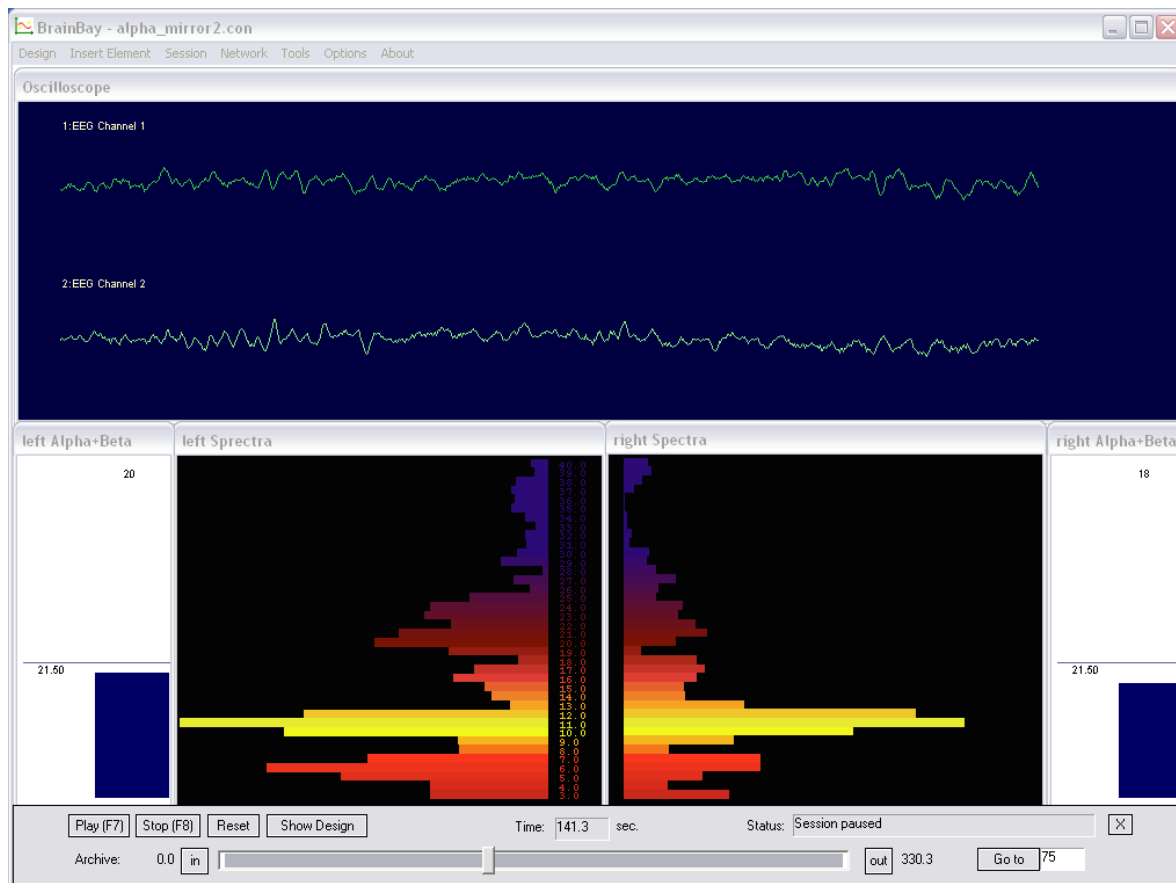


6. Example Designs

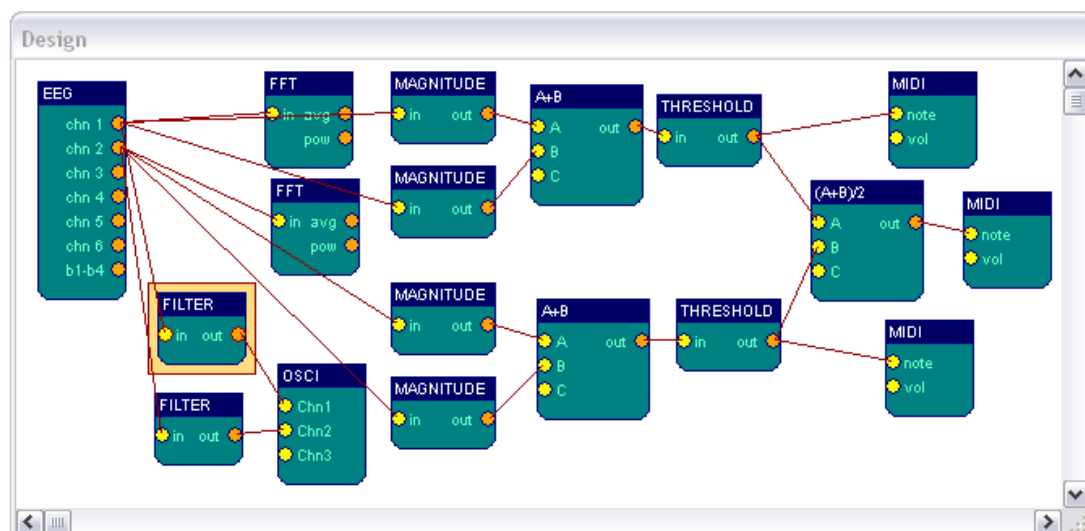
The following examples will show practical applications of design elements. Three designs will be described briefly, one for neurofeedback, one for muscle feedback and a head-mouse design.

6.1 A design for Alpha / Beta – Neurofeedback

The following design shows realtime-processing of a two channel monopolar EEG recording. Three Silver Electrodes were mounted at C3, C4 and Cz - positions according to the 10/20 - system. The reference point was Cz. An abrasive conductive gel was used to improve electrode-skin contact. The impedance of the electrode-skin-connection was about 10kOhm. Sampling was performed by the MonolithEEG amplifier at 256 Hz, using the P2 firmware protocol and a baudrate of 56700 bits/sec.

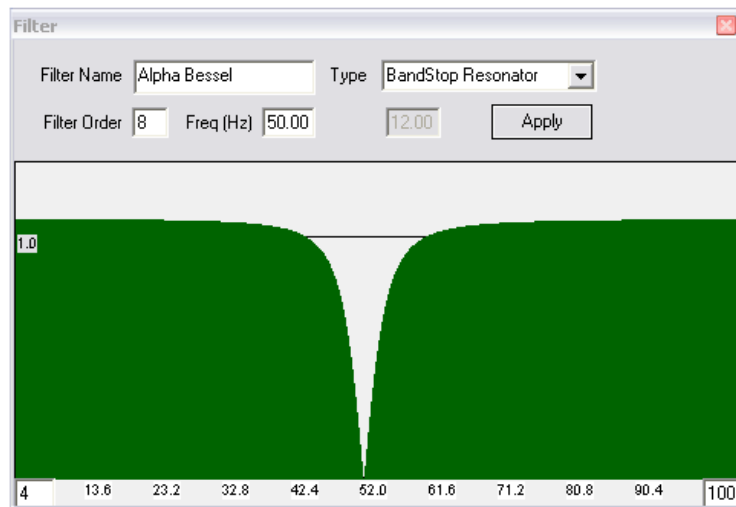


The oscilloscope window shows the EEG signals which have been 50Hz - filtered. Two FFT - spectrum analysers show the frequency spectra of the EEG signals (brain-mirror display). Magnitudes in alpha- and beta range give audio feedback via multiple midi channels. The two meter-windows show current magnitudes and thresholds for the audio-output.

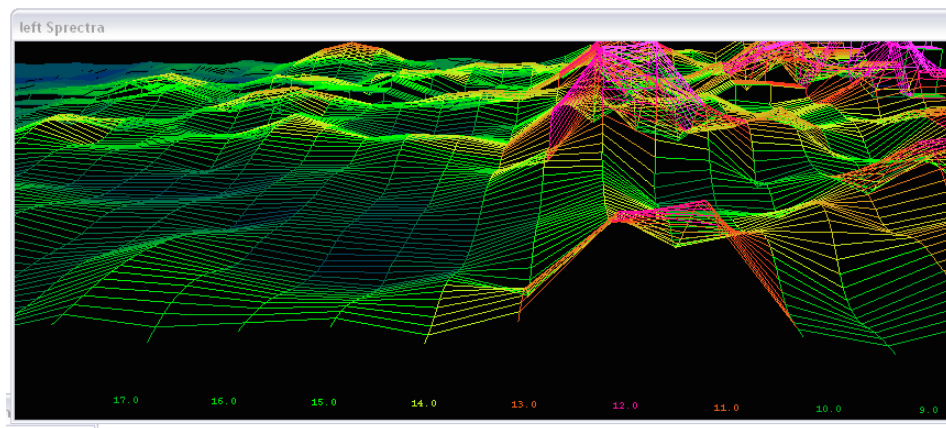


The design consists of the following elements (per channel):

- a **filter** of the type 'Bandstop Resonator', Bandstop-frequency=50 Hz, Order=8
- a connection to the **oscilloscope**, to display the filtered EEG signal
- a **FFT** spectral display, calculation every 10 samples, gain 250, 128 Bins, display range 3 - 40 Hz, style: bar-graph, user-edited color palette
- two **magnitude** elements, using butterworth - bandpass filters in the range of 8-12 Hz and 16-20 Hz, filter order 4, gain 100 %
- an expression **evaluator** that adds the two calculated magnitudes
- a **threshold** element, averager set to 223 samples, threshold set to > 21,5 uV
- a **midi** tone generator, using a user-defined harmonic scale and an output interval of 70 samples.



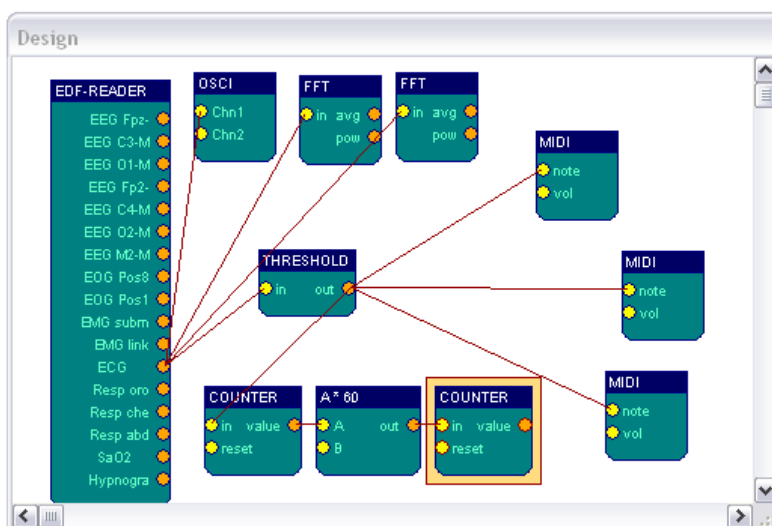
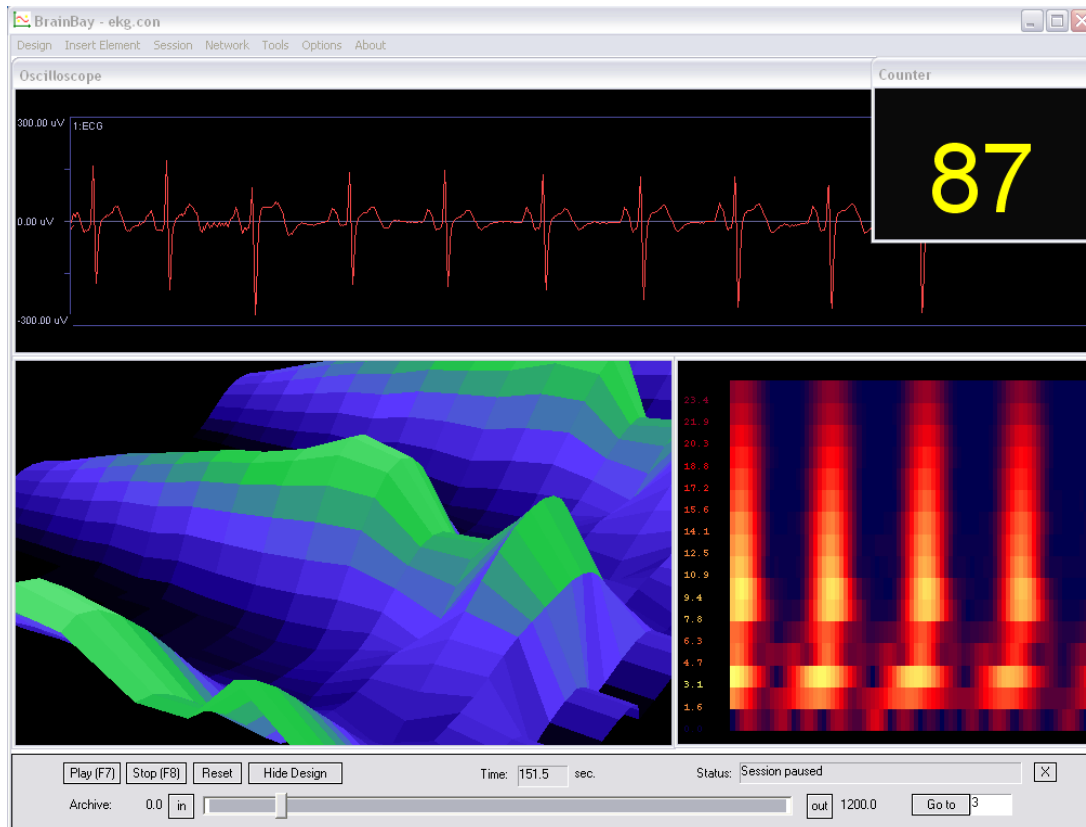
above: the frequency - response of the 50Hz - bandstop filter



the *Style* parameter of the FFT-element was changed from *bar-graph* to *3d line-grid*

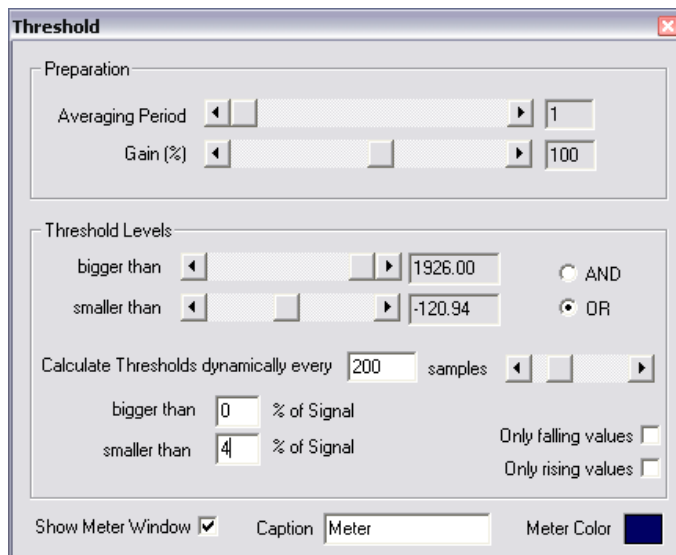
6.2 Heartbeat-Feedback from a multichannel EDF-recording

The following extracts the heartrate from a 17-channel multimodal biosignal recording stored in an EDF-file. ECG and frequency-spectra of the contraction of the heart muscle are displayed. For calculating the heartrate, a dynamic threshold level is applied. Acoustic feedback for the heartbeat is created by three midi-generators.



The ECG-output of the EDF-file reader source is connected to display- and processing elements. The first FFT-display is in *3d*-mode, the second in *moving spectrogram*-mode. Both FFTs use 32 bins and a calculation interval of 4 to enable fast displays. The threshold-element detects the R-wave of the QRS-complex.

The heartrate is calculated by a counter-element which counts the frequency of threshold-activation. The output of the counter (given in Hz) is multiplied by 60 using an expression-evaluator to get beats per minute. The second counter is used only as a display for the value.



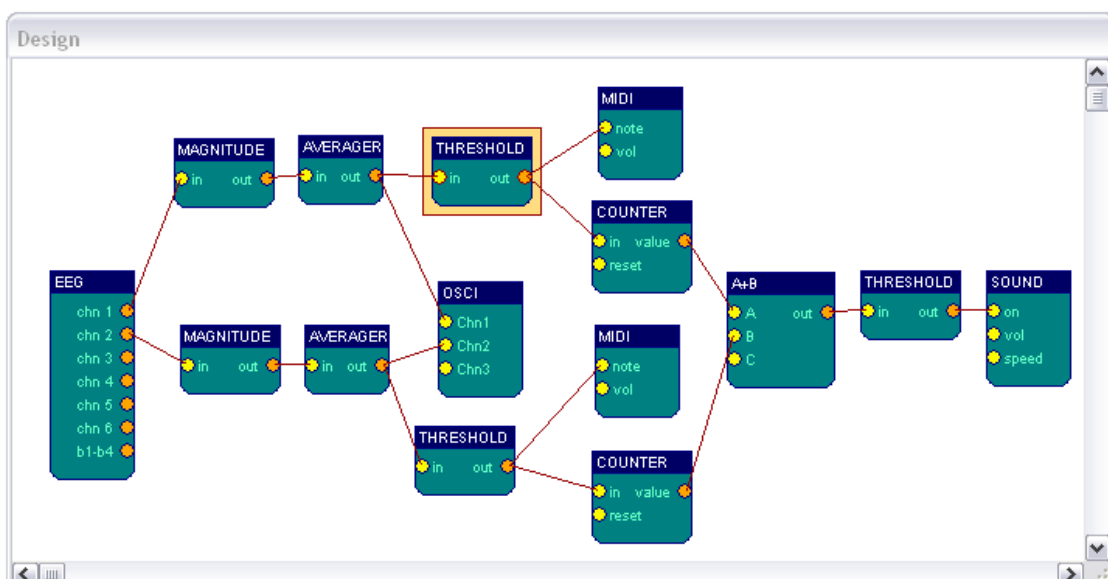
R-Wave-detection:

The R-wave in this ECG- recording is characterized by a large negative spike. This peak can be used to detect one heart-beat with a threshold element. A way to adapt to the varying amplitude of the heart-signal is to use a dynamic threshold: the level to pass values is set to 4% of all signal values in the last 200 samples.

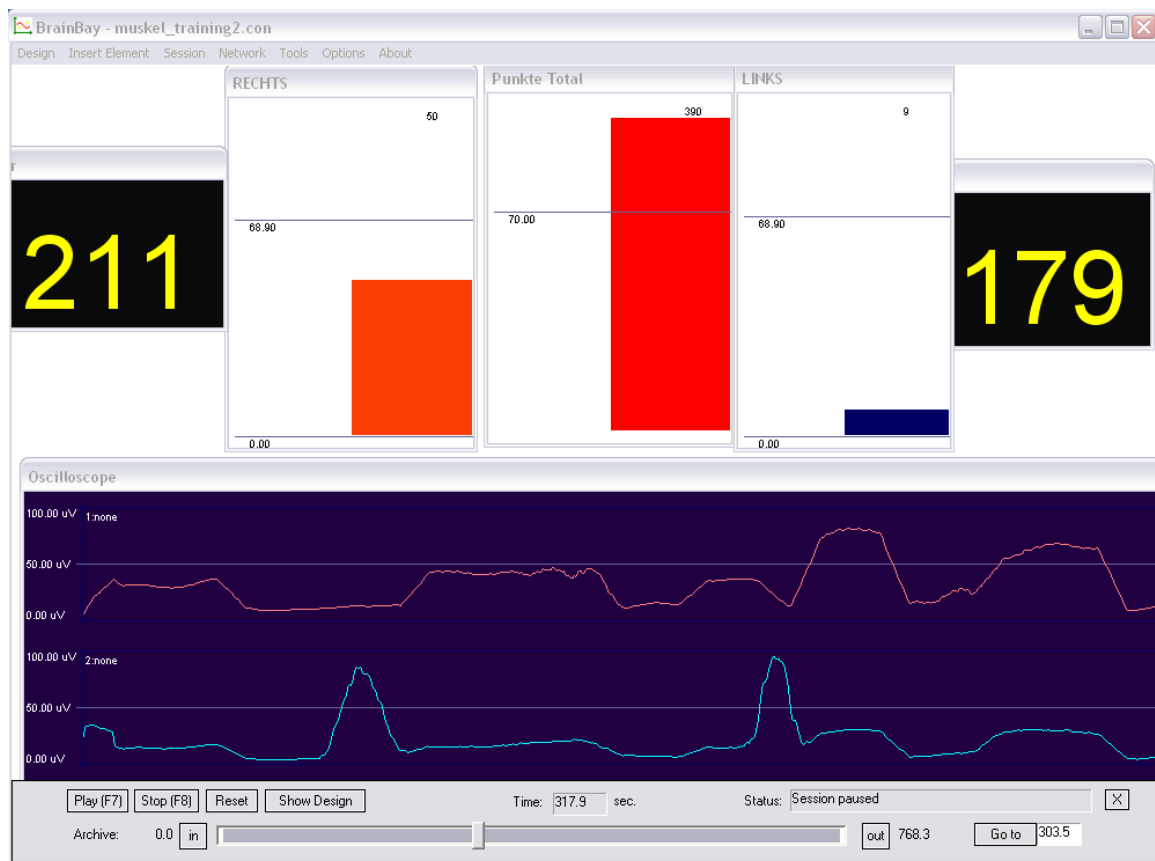
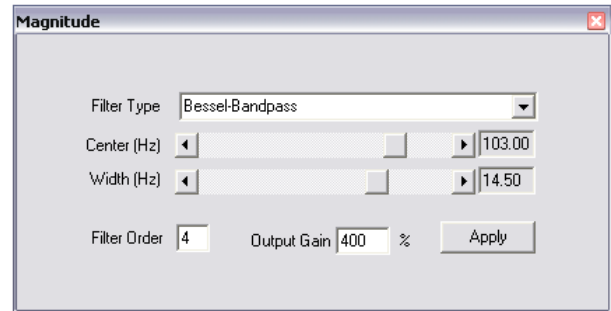
For audio feedback, three midi-generators with the same type of harmonic tone-scale were used. As the detected peak of the R-wave can be very short, a fast tone output-interval is needed to recognize each heartbeat. Therefore, the output-intervals for the midi-elements were set to 1, 2 and 3, which corresponds to 100, 50 and 33 tones per second. (the sampling rate for this EDF-recording is 100Hz). The different intervals for tone-generation lead to slightly different notes which fit together because of the harmonic scale.

6.3 A design for Muscle-Feedback and –Rehabilitation

The following design can be used to perform muscle feedback for training and rehabilitation purposes. The example recording was done with a MonolithEEG amplifier running in P21-bidirectional mode and at 600 Hz sampling frequency. The amplification has been set to a low level, corresponding to high muscle potentials of about 500 uV. Muscle activity could be measured using standard *BlueSensor* EMG surface-electrodes attached to the left and right thigh. Four electrodes used in bipolar mode give a two-channel setting.



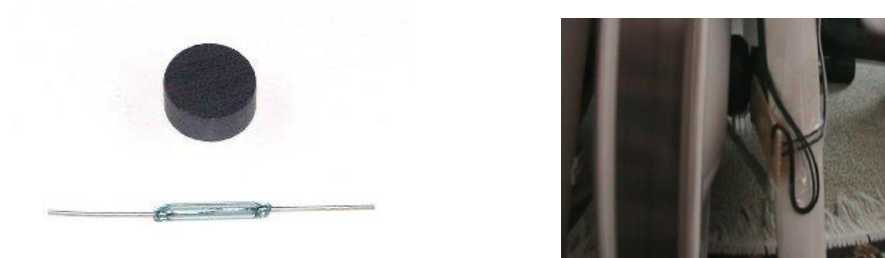
To get the muscle contractions and filter out lower-frequency noise, one magnitude-element per channel is used. The parameters are set to a 4th-order bessel bandpass filter of 85-115 Hz range. The magnitude output is averaged for a period of 300 samples, (half a second at 600 Hz sampling frequency). Thus, the fast and jumpy magnitude-value is made smoother and can be used as input for the threshold-element.



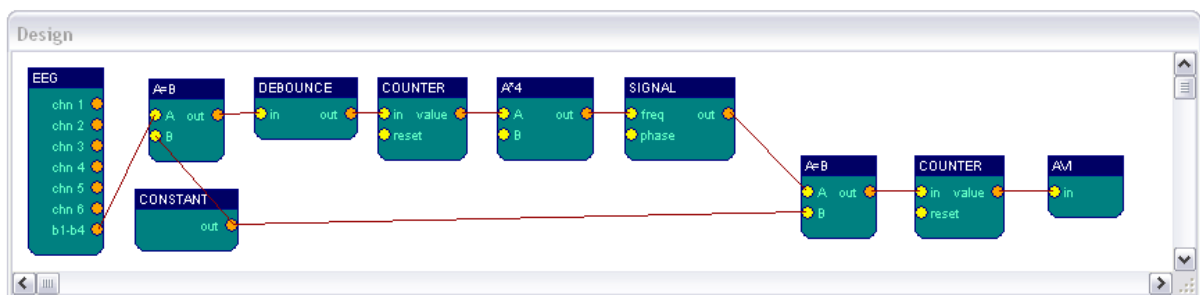
EMG-activity of two locations can be seen on the meter-displays. An acoustic signal rewards the user when a selected level of muscle-contraction has been reached. Two counters show how often the muscles have been contracted in total. When a certain number of contractions has been performed, a sound will be played that indicates the completion of a training-session.

6.4 Controlling a movie using an external Reed-Switch

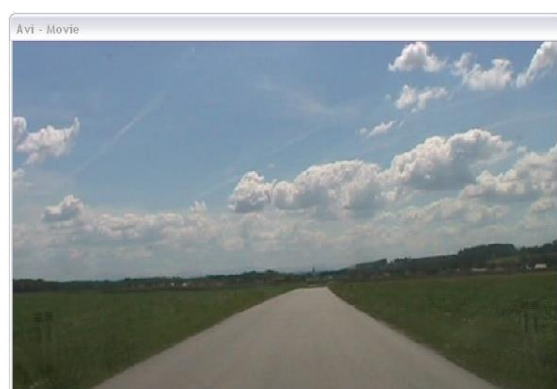
The following design illustrates how to use an external switch to control the frame-rate of a movie-playback. As an application of this design, a bicycle home-trainer could easily be turned into a multimedia recreation device: A magnet mounted on the turning wheel closes a reed-switch when passing; the design transforms the switching information to a desired playback speed of the movie.



Above figures show the reed-switch & magnet and mounting of the switch on the home trainer. The switch is connected to a digital input port of the ModularEEG (4 input ports are reserved for buttons/switches). The switch state is present at the *B1-B4* - output port of the *EEG amplifier* element.



In the design, the *b1-b4* output is compared with a constant value (in this case, button 2 has been connected and value 11 indicates a switching of the reed contact). The debounce-element prevents double- or triple-impulses when the magnet passes. The first *counter* measures the frequency of the switching pulse. Its output controls the frequency of a signal generator. Thus, the switching speed of about 1-4 Hz coming from the wheel/magnet is transformed into a frame pulse for the movie of about 5-30 frames per second. The current frame number is calculated by the second *counter* element and fed into the avi player element.



6.5 A Camera - Mouse application with clicking functions

This design example shows how to use the camera-element to control the mouse cursor and perform clicking- actions by head movements. With the option *feature-tracking* enabled, the *camera* - element tries to detect the user's face and extract two positions:

The nose is marked with a green circle and the chin is marked with a yellow circle.

When red circles appear, the tracking algorithm has lost the face and a re-initialisation of the face-tracking will be performed.

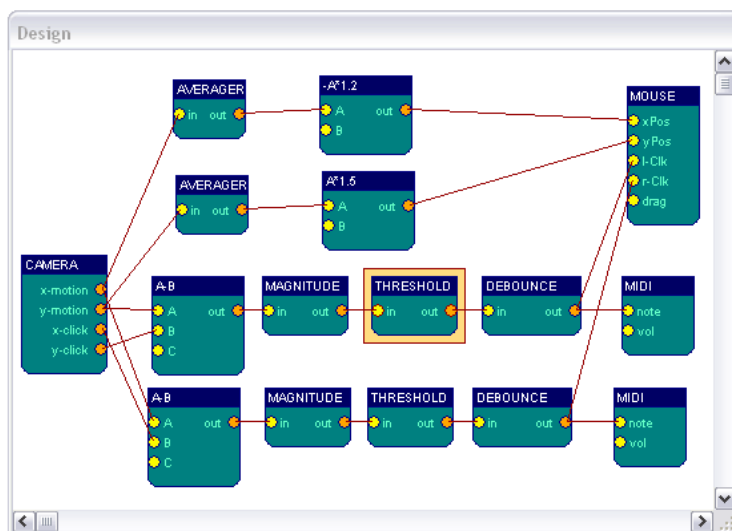
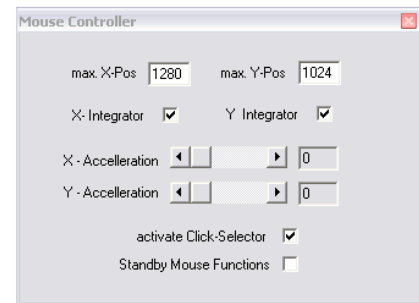


The outputs of the camera element:

x-/y-motion = relative changes of nose-position in pixels

x-/y-click = relative changes of chin-position in pixels

This information is transformed to fit the mouse controller element. Cursor position is set by the *xPos* and *yPos* inputs of the *mouse* - element. The nose-movement can serve these inputs, after being averaged (5-10 samples, to prevent jittering) and scaled to get the desired speed, x-direction is mirrored to fit left and right movement.



To issue left and right clicks, relative changes in the nose-chin distance are calculated by subtracting *x-click* from *x-motion* and *y-click* from *y-motion*: moving the head up and down or left or right changes nose and chin positions in nearly the same way, the difference will be near zero. Moving the chin up and down shows up in the y- difference, left and right chin movements are reflected in the x- difference. Two *magnitude* - elements with a passband from 3.5 -5.5 Hz indicate rapid chin movements in up/down or left/right direction. *Threshold* - elements set a level for the minimum magnitude to generate a mouse-click. The *debounce* - element prevents from double- or triple-clicks where they are not wanted. The clicking information is fed into the *l-clk* and *r-clk* input ports of the mouse element.